

Strategic Environmental Assessment for Linköping Municipality

Municipal level assessment of Linköping masterplan - Land Use Sector

A.Y. 2025/2026

AE2507 Group 8



Giuseppe Todaro

Emilian Tietze

Srishti Dasgupta

Siri Åsman

Frida Yousef Lahdo

Zexuan Zhang

Table of Content

1. Background	4
1.1 Introduction & Aim	4
1.2 Strategic Environmental Assessment (SEA)	6
1.2.1 Introduction to SEA	6
1.2.2 SEA and the Planning Process	7
1.2.3 Types of SEAs	8
1.2.3.1 Conceptual Approach to SEA in This Report	9
1.3 Boundaries and limits	11
2. Methodology	12
3. Result of the assessment	14
3.1 Description of the sector	14
3.1.1. Land-use sustainability issues	15
3.1.2. SWOT and crossed-SWOT analyses for the land-use sector	15
3.2 The Linköping Masterplan	19
3.2.1 Description of the plan	19
3.2.2 The importance of planning	20
3.3 SEA report components	21
3.3.1 Strategic Reference Framework (SRF)	21
3.3.1.1. SEA Process	21
3.3.1.2. Scope	22
3.3.1.3 Relevant International and National Legislation and Policy Framework	23
3.3.2 Governance framework and stakeholders	24
3.3.2.1 Public sector	26
3.3.2.2 Private sector	26
3.3.2.3 Civil society	26
3.3.2.4 Academia	26
3.3.3 Results from Interviews	27
3.3.3.1 Living and affordability	27
3.3.3.2 Transportation and job	27
3.3.3.3 Environment	28
3.3.3.4 Development and restrictions	28
3.3.3.5 Identified challenges	28
3.3.3.6 Current framework	29
3.3.3.7 Sustainability	29
3.3.3.8 Growth and direction	29
3.3.3.9 Stakeholders	29
3.3.4 Problem statements and SEA elements: from CDFs to indicators	30
3.3.5 Assessment criteria and indicators	32
3.3.5.1. H: Cultural Heritage	32

3.3.5.2. I: Industry renovation	33
3.3.5.3. U: University and scientific development	34
3.3.5.4. E: Ecosystems and urban ecology	34
3.3.5.5. S: Civil society and a vibrant city	35
3.3.6 Strategic Options	37
3.3.6.1. SO I. Heritage-led development	38
3.3.6.2. SO II. Cultural Tourism	40
3.3.6.3. SO III. Innovative Industry	41
3.3.6.4. SO IV Sustainable Green Industry	42
3.3.6.5. SO V. Campus with zero net-carbon emission	43
3.3.6.6. SO VI. Affordable on campus student housing	43
3.3.6.7. SO VII. Ecology and green city	44
3.3.6.8. SO VIII. Transport	45
3.3.6.9. SO IX. Housing	46
3.3.7 Opportunities and risks	47
3.3.8 Final matrix	48
3.3.9 Monitoring and Follow-up	49
4. Discussion and conclusion	53
4.1 Discussion recommendations	53
4.2 Uncertainties and gaps	54
4.3 Conclusion	55
5. References	57
6. List of Figures	64
7. List of Tables	64
Appendices	65
A - Inhabitants interview (Anna Brobeck & Linnea Ingren)	65
B - Municipal comprehensive planner interview (Mikael Sonesson)	70

1. Background

This document serves as a policy and plan-level Strategic Environmental Assessment (SEA) report, primarily aiming to evaluate the long-term spatial strategies outlined in the Linköping Masterplan. Its primary contribution is to provide a structured, evidence-based framework that guides the city's development trajectory, ensuring that the urgent demand for housing and infrastructure does not compromise long-term ecological resilience or cultural heritage. The intended audience for this report includes municipal planners, local policymakers, regional authorities such as the County Administrative Board, and engaged citizens who participate in the city's democratic planning processes. Ultimately, this report is critically needed because Linköping is experiencing rapid population growth toward 200,000 inhabitants and facing major infrastructure shifts, such as the Ostlänken high-speed railway. Without a proactive SEA to balance competing land-use demands, the municipality risks irreversible consequences, including the loss of valuable arable land, fragmentation of green corridors, and unchecked urban sprawl.

In order to ensure that these challenges are addressed in a coherent and forward-looking manner, the SEA must be carried out by an independent and interdisciplinary consultant team with the capacity to integrate environmental, spatial, social, and economic perspectives. The consultant team acts as an independent, interdisciplinary group of experts responsible for conducting the SEA of the Linköping Masterplan. Our role is to provide objective, evidence-based analysis of environmental, social, and spatial impacts, assess reasonable alternatives, and propose mitigation measures. We ensure compliance with the SEA requirements under the European Union Directive and Swedish environmental legislation, while supporting transparent and informed decision-making.

The team has been appointed by the Municipality of Linköping to independently evaluate the long-term sustainability implications of specific strategic choices. The appointment ensures professional neutrality and strengthens the credibility of the assessment.

1.1 Introduction & Aim

Linköping Municipality, located in the heart of the Östergötland county, stands as one of Sweden's fastest-growing cities and serves as a critical engine for regional development (Linköpings kommun, 2010). Historically known as a cathedral and administrative city, Linköping has transformed into a modern hub for innovation, often branded as the "Aviation Capital of Sweden" due to its strong aerospace heritage with SAAB and its dynamic high-tech industry centered around Mjärdevi Science Park. Geographically, the city is uniquely situated where the forested landscapes of the south meet the vast, fertile agricultural plains of Östgötaslätten to the north. This location provides the city with a distinct character defined by the river Stångån and Kinda Canal, but it also presents a complex planning challenge: managing urban expansion without compromising some of Sweden's most valuable agricultural land.

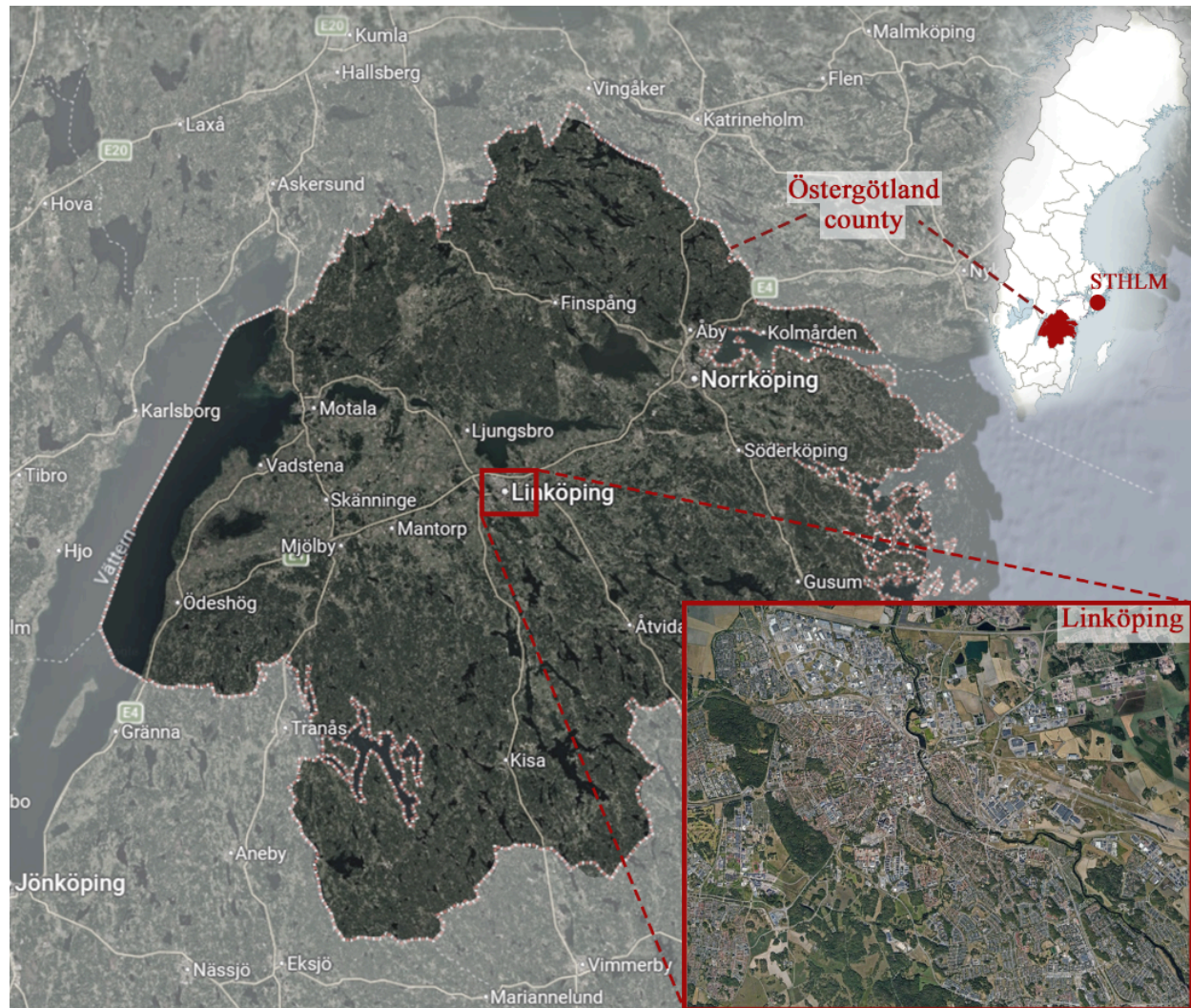


Figure 1: Insertion of the city of Linköping within the general framework of Östergötland County, which in turn is situated within the mosaic of Swedish counties. Source: authors elaboration through Adobe photoshop. Google satellite base.

The municipality's current Comprehensive Plan (Översiktsplan) is driven by an ambitious vision to transform Linköping into a "vibrant and attractive city where dreams become reality", while simultaneously respecting nature and historical heritage (Linköpings kommun, 2010). To accommodate a projected population growth toward 170,000 inhabitants and beyond, the plan abandons the historical pattern of sprawling outward in "wedges". Instead, it adopts a strategy of creating a "denser, rounder, and more cohesive city". This vision relies heavily on strategic densification and inward growth to minimize land consumption and transport distances. A central catalyst for this transformation is the planned Ostlänken high-speed railway and the relocation of the travel center (Resecentrum), which is expected to shift the city's center of gravity and unlock new districts like Kallerstad for mixed-use development (see Figures 2 and 3).



Figure 2: Linköping Resecentrum (Travel Center)



Figure 3: Planned Ostlänken (The East Link) from Järna to Linköping -2035 (Trafikverket, n.d.)

In this project, our group applies a Strategic Environmental Assessment (SEA) specifically to the land use sector of this comprehensive plan. The application of SEA in this context is critical, as the municipality's strategy to densify and expand major infrastructure carries significant environmental implications. Our analysis focuses on evaluating how the plan's land use designations interact with environmental sustainability goals, specifically assessing the trade-offs between satisfying the urgent demand for affordable housing and infrastructure against the irreversible loss of arable land and the fragmentation of green corridors (Linköpings kommun, 2010). By scrutinizing these strategic choices, this SEA aims to determine if the current land use plan effectively balances Linköping's aggressive growth targets with its commitment to long-term ecological resilience.

1.2 Strategic Environmental Assessment (SEA)

1.2.1 Introduction to SEA

This project will, thereby, analyze the comprehensive plan of Linköping in relation to the land-use sector. The analysis will be conducted using a perspective of strategic thinking within Strategic Environmental Assessment (SEA). A SEA is a process applied to ensure sustainability and environmental consideration within development on local to national levels. It functions as a framework for strategic decision-making (Therivel, 2010), primarily targeting policies, plans, and programmes rather than individual projects. In doing so, it supports sustainable development by fostering interdisciplinary engagement in planning processes and by anticipating and mitigating adverse environmental impacts (Therivel, 2010).

Given its flexibility, SEA can be applied across sectors and governance levels. Its procedural structure allows environmental considerations to inform multiple stages of planning and decision-making (Therivel, 2010). Beyond impact identification, it provides strategic direction by embedding sustainability objectives into policy design and by formulating implementation guidance aligned with environmental values (Partidário, 2012).

SEA is thereby one of the central sustainability instruments for planning and decision-making processes. Another central tool is Environmental Impact Assessment (EIA). These two tools, SEA and EIA, are complementary but different. The most significant difference is the stage at which one is applied. SEA is initiated much earlier and within a much higher level of decision-making considering larger and more long-term impacts, compared to EIAs (Therivel, 2010). Hence, an SEA aims to address the larger strategic thinking and actions that determines the trajectory of which projects will be assessed by an EIA, making EIA more project based compared to SEA. Withal, an SEA analysis will be applied to the land use sector within Linköping.

1.2.2 SEA and the Planning Process

The purpose of Strategic Environmental Assessment is its inclusion as a component in the planning process, rather than as a distinct and/or subsequent evaluation. It is conducted concurrently as opposed to sequentially, integrating the procedures of formulation of policies, plans, and programs rather than as a standalone activity, such that the environment influences decision-making at every stage (Therivel, 2010; Partidário, 2012). In land-use planning, such strategic decisions are crucial, as the allocation of land for urban development, for instance, can have a long-lasting irreversible impact on the environment. SEA thus plays a key role as a preventive tool (Sadler et al., 2011).

In municipal planning, SEA plays a critical role in facilitating comprehensive plans in which the best way to integrate environmental goals into municipal planning is highlighted. In Sweden, comprehensive plans are used for long-term planning of land use development; they integrate sectoral interests like housing, transport, climate change, and ecosystem services.

In European Union law it is stated that an environmental assessment is to be carried out for certain plans and programs which are likely to have significant effects on the environment. SEA supports this process by providing a clear identification of relevant environmental issues, a consideration of cumulative effects, prediction of long-term consequences, and an evaluation of reasonably anticipated alternatives (European Commission, 2001). This allows SEA to be positioned as a link between upper-level goals, like regional or even country-level environmental goals, and local planning goals.

Moreover, SEA enhances governance and transparency in the planning process by fostering consultation and interdisciplinary coordination, since the SEA framework brings together planners, environmental authorities, and other relevant stakeholders in the process. This ensures that the legitimacy of the planning process is improved (Partidario, 2012). Thus, while other environmental management systems seek to establish whether an action can be supported or otherwise, SEA provides an evidenced knowledge base for decision-making and accountability (Therivel, 2010).

1.2.3 Types of SEAs

In the land-use Sector, the type of SEA applied depends on the objectives of the project. For the purposes of this report, SEA should be implemented within the policies. The SEA should guide Linköpings municipality's land-use sector to ensure that considerations for the environment are included in the early stages of decision-making. The assessment should begin with screening, which is a tool used to examine whether the proposed policies, plans, or programs for the proposed project would result in significant environmental impacts, therefore, a full SEA is necessary (European Commission, 2023). Following screening, the process then turns into scoping, which, during this process, key environmental issues are identified. Scoping in the SEA highlights relevant concerns. The SEA addresses these concerns through the planning process (European Commission, 2023).

Before implementing the SEA, it must act in accordance with the EU SEA Directive and possibly align with the Sustainable Development Goals (SDG's). The SDGs were established to eliminate today's world issues involving poverty, peace, justice, and protecting the planet. The legal and institutional framework for SEA in Sweden is established by the environmental code (1998:808) and the ordinance on environmental assessment of plans and programs (1998:905). These regulations make sure that certain plans and programs undergo SEA, furthermore, it also includes environmental impact assessments, with additional guidance provided in the ordinance. Municipalities, for example Linköpings municipality, are responsible for applying SEA at the local level, consulting relevant authorities, and the county administrative board with chapter 6, section 13 of the environmental code. These laws should also support regional and national standards (European union, 2023). For the land-use sector in Linköpings Municipality, the most relevant goals include SDG 3 (Good health and well-being), SDG 13 (Climate Action), and SDG 15 (Life on land) (United Nation Development Program, u.d.). In Sweden, SEA is conducted under *Miljöbedömning av planer och program*, as directed by the Swedish environmental protection agency (2023). The different types of SEA concern policy, plan, and program levels.

At the policy level, SEA guides large scaled land-use policies, providing a strategic framework. The land-use policies are implemented to ensure environmental integration prior to decision-making. According to UNECE (2023), policy level SEA is structured as a process created for evaluation of policies, plans, and programs for the potential impacts that the environment proposes. This structure distributes the focus, rather than solely relying on plans or projects. It is primarily abstract, focusing on long-term sustainability objectives and cross-sectoral effect, while providing a linear approach to decision-making.

At the plan level, SEA transcribes these broad policy objectives. It compresses them into spatial and sector specific guidance, applying to understandable municipal plans, such as the comprehensive plan (översiktsplanen). The plan level contains programs affecting residents, including housing and rural development (European commission, 2023). This report, in contrast to plan level SEA, examines Linköping Municipality's comprehensive plan. The purpose of analysing the comprehensive plan is to collect understanding of the land use sector and its future developments in Linköping. The plan-level provides a detailed environmental assessment than the policy SEA, ensuring that plans are consistent within the sustainable objectives.

At the program level, SEA becomes more detailed, and targeted policies, such as housing development or specific initiatives that Linköping may plan. The program level is more detailed than the policy-level, still strategic and supports future development in accordance with legislative and regulatory frameworks (Naturvårdsverket, 2023). To achieve a program level SEA, interviews with the public will be held. It is therefore important that, throughout the SEA process, that stakeholders and public engagement are prioritized. In the context of Linköping, it is therefore relevant to involve a consultation between inhabitants, authorities, as well as including the county administrative board. This is to ensure that environmental concerns are included with transparency. Lastly, monitoring and follow ups are set to complete the SEA cycle, supporting the outcomes of the land-use sector decisions over time to inform future planning and improve sustainable practices.

1.2.3.1 Conceptual Approach to SEA in This Report

SEA can be conceptualised in two fundamentally different ways (Azcárate, 2026):

- (1) as an extension of Environmental Impact Assessment (EIA) procedures to higher decision levels; and
- (2) as a strategic, integrative framework embedded within policy and planning processes.

These two approaches differ in purpose, timing, methodology, and influence on decision-making.

EIA-based SEA applies the logic and methodology of project assessment to plans and programmes. It typically follows a linear structure: establishing a baseline, predicting impacts of a defined proposal, identifying mitigation measures, and compiling an environmental report see Figure 4. In this model, the assessment tends to evaluate the environmental consequences of a largely predetermined spatial strategy (Azcárate, 2026). While such an approach can improve transparency and ensure that environmental effects are documented, it often remains reactive. Environmental concerns are treated as constraints to be managed, rather than as structuring principles that shape the development strategy itself.

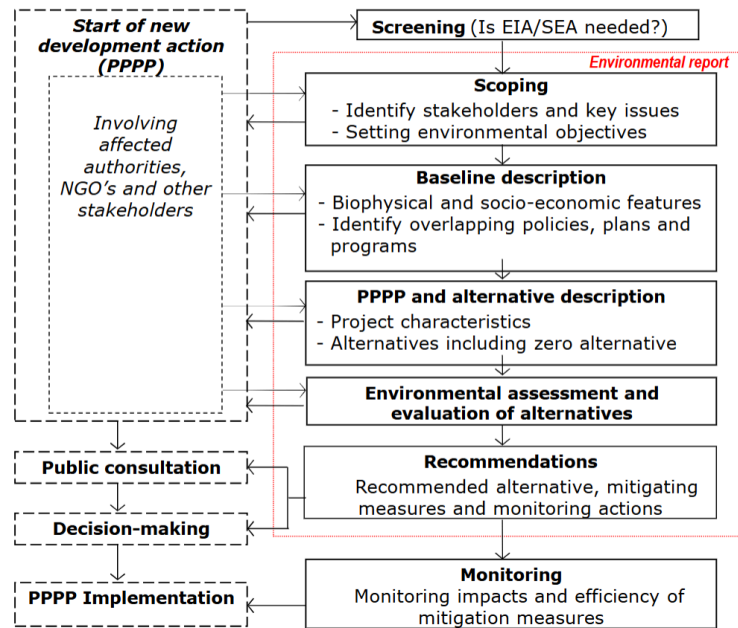


Figure 4: EIA Based SEA process Source: (Azcarate, 2026)

In contrast, strategic thinking-based SEA (As demonstrated in figure 5) reconceptualises environmental assessment as an integral component of strategic decision-making. Instead of impacts, it interrogates which development trajectory is most consistent with long-term sustainability objectives. This approach is embedded in the early stages of plan formulation and operates iteratively alongside the planning process. It emphasises the comparison of alternatives, the identification of cumulative and indirect effects, and the alignment of spatial strategies with environmental objectives.

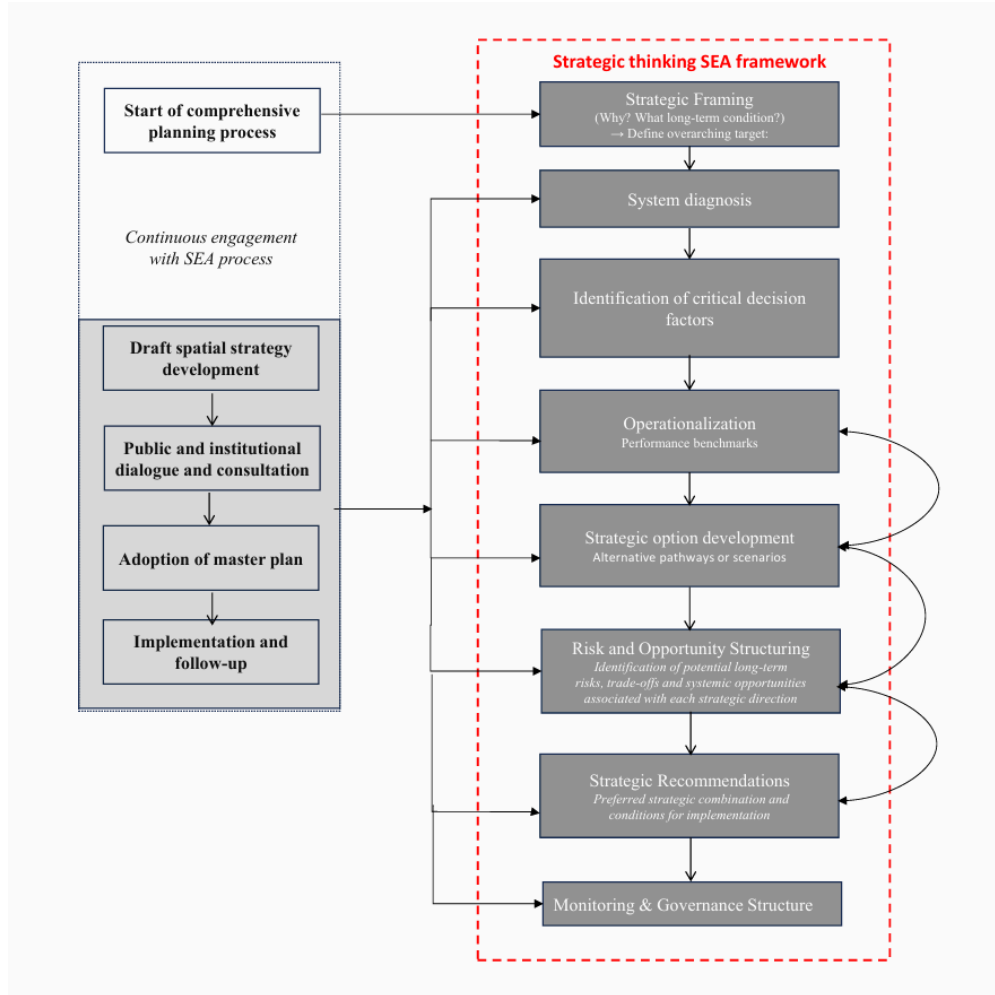


Figure 5: Flow diagram showing Strategic thinking SEA process in planning | Source: Authors, adapted from (Azcárate, 2026)

1.3 Boundaries and limits

The following work investigates the town of Linköping, analysing the urban centre at the municipality level. As mentioned before, the study concerns the assessment of the Linköping overall masterplan, in particular attentioning the aspects of the document relying on the land-use sector.

There are several delimitations to the aforementioned analysis. The first obstacle is represented by the language itself. The Linköping Masterplan, which is completely in Swedish, has been translated into English through an automatic translator, which could make cumbersome specific passages in reading the document. Another important limitation is represented by the available time span, which is limited and proportioned to an academic exercise. A broader time arch would be necessary for the development of a more complete and thorough analysis. Moreover, the lack of data and the impossibility to reach specific stakeholders represented a critical impediment for the whole study. Despite these obstacles, the current work aims to conduct a holistic and careful assessment of the Linköping Masterplan.

2. Methodology

For this Strategic Environmental Assessment (SEA) of the land-use sector in the Linköping municipality, we chose a qualitative and cyclical research design involving literature studies, interviews, lecture content, and structured group discussions. With a focus on early problem formulation and iterative redefining of practices, this project followed a customized SEA procedure. The approach is qualitative, iterative, and aligned with strategic SEA theory as described by Therivel (2010) and Partidário (2012).

To start things off a systematic analysis of planning and policy documents to identify the scope and context of the land-use sector was conducted. The main documents included the comprehensive plan of Linköping Municipality (Linköpings kommun, 2010), Swedish planning acts according to the Planning and Building Act (Plan- och bygglag) (2010:900), and the Environmental Code (Miljöbalken) (998:808). Moreover, the European frameworks of the SEA Directive (*European Commission, 2001*) and the LULUCF Regulation were considered to understand the context of land use in climate change mitigation and environmental management. Partidário (2012), Therivel (2010), and the Brundtland Report (WCED, 1987), forming the theoretical basis for the sustainability goals and the assessment process were utilized to assess a framework of SEA procedure. The document analysis helped to identify the sustainability issues of land consumption, biodiversity conservation, agricultural land conservation, urban densification, and climate change adaptation. The analysis also helped to understand the institutional framework of the Swedish planning system. Based on this contextual understanding, a stakeholder mapping exercise was carried out. The stakeholders were identified through an analysis of the comprehensive plan, statutory consultation processes (samråd), and group brainstorming sessions. The stakeholders were classified into public authorities, private sector, civil society, and academia. Their level of influence and interest in land use was qualitatively determined, which would form the basis of a governance framework to be discussed later in this report. This stakeholder mapping also helped in identifying participants for the interviews.

Two interviews were carried out to supplement document analysis with experience and practice-based knowledge. We approached different institutions such as SAAB and the university of Linköping for a possible interview but got rejected for unknown reasons. Two interviewees were former students of urban and regional planning at Linköping University and have lived in Linköping for about three years. An interview with these students was conducted in a semi-structured way while interviewing both students at the same time. The other interviewee was a comprehensive planner of Linköpings municipality who has lived in Linköping for six years. This interview was done via email. Interviewees were chosen for their combined experience of the city and knowledge of planning processes. Considering the short duration of the project, the choice was made to strike a balance between accessibility and relevance to the land-use sector.

An interview guide was prepared based on topics identified from the literature review and preliminary SWOT analysis. The guide covered views on urban structure, the availability and affordability of housing, mobility networks, accessibility to green spaces, harmony of the community, and perceived strengths and weaknesses in the

ongoing urban development. This guide was provided to the interviewees in advance so they could prepare themselves for the questions. The “in-person” interview was carried out in a semi-structured manner meaning that the interview guide was followed but based on the answers new questions were formulated on the spot, to enable comparison while still being flexible to pursue new themes. It lasted for about 50 minutes. During the meeting a dictation program (I-phone notes) was applied, which transcribed the whole interview automatically. The resulting notes have been successively adjusted and fixed in order to obtain a more polished version. The second interview with the municipality of Linköping was conducted via email. Both interviews can be found in the appendix.

The internal group discussions and brainstorming activities were core to the analysis process. The SWOT analysis was done through a two-stage process where each member of the group initially identified the strengths, weaknesses, opportunities, and threats individually through literature. These were then collectively discussed and further developed into a collective matrix. The cross-SWOT analysis helped to further integrate the internal elements (strengths and weaknesses) of the group with the external dynamics (opportunities and threats) to facilitate the development of strategic directions. This process was particularly crucial in advancing from the descriptive analysis to the strategic process, in line with SEA as conceptualized by Partidário (2012).

Based on the SWOT and cross-SWOT analysis, we developed strategic objectives that were in line with the sustainability principles as enunciated in the Brundtland Report (WCED, 1987) and EU environmental policy. The Critical Decision Factors (CDFs) were then identified as the leverage points in the land-use system where strategic action could make a significant difference to the environmental and social outcomes. Each CDF was further developed through qualitative assessment criteria to ensure consistency between analysis and evaluation. Strategic decisions were then developed and evaluated using a qualitative risk-opportunity matrix. The decisions were examined in the context of the identified CDFs and sustainability targets, based on knowledge from the literature, interviews, and group brainstorming sessions. This allowed the SEA to be a learning and strategy development process rather than a static evaluation.

Finally, monitoring concerns were identified from the risks and opportunities, with emphasis on climate accounting in the context of the LULUCF Regulation and Swedish environmental governance structures. While there was no quantitative GIS analysis in the context of this study, the notion of spatial monitoring indicators was developed to monitor land change, ecological connectivity, and urbanization across space and time.

There are a number of methodological limitations that must be recognized. The research was conducted over the course of an academic period, which imposed a natural limit on the number of interviews that could be conducted. Only three interviewees were actually interviewed, which imposed a limit on the scope of opinions that could be taken into account. In addition, the research was essentially qualitative in nature and did not involve any quantitative modeling or spatial analysis. However, the application of methodological triangulation, which involved document analysis, stakeholder mapping, interviews, and structured group analysis, served to improve the internal validity of the findings. The methodology applied is essentially strategic, qualitative, and participatory in nature, and is consistent with the established theory of SEA (Partidário, 2012; Therivel, 2010).

3. Result of the assessment

3.1 Description of the sector

Land use is a foundational sector that underpins food systems, housing, infrastructure, economic production, and the provision of ecosystem services. It structures how space is allocated between agriculture, forestry, settlements, industry, transport, and natural habitats, thereby directly shaping both human development and ecological functioning.

At the same time, land use is a major driver of environmental pressure. Processes such as land take, soil sealing, deforestation, peatland drainage, intensive agriculture, and urban expansion contribute to soil degradation, biodiversity loss, water stress, and greenhouse gas emissions (*Land use sector; n.d.*). These impacts are often cumulative, spatially fixed, and difficult or impossible to reverse.

The sector also plays a dual role within climate policy. Land can act both as a source of emissions, through activities such as agriculture, land use change, and forest clearance, and as a carbon sink, through forests, wetlands, soils, and harvested wood products (*Land use sector; n.d.*). This duality makes land management a critical lever for both mitigation and adaptation strategies.

Within the European Union, this role is formalised through the integration of the land use, land use change, and forestry (LULUCF) sector into the climate framework. Regulation 2018/841 establishes binding targets for Member States to maintain and enhance carbon sinks (*European Parliament & Council, 2018*). The 2021–2025 “no-debit” rule requires that emissions do not exceed removals, transitioning to a collective Union target of 310 million tonnes of CO₂ equivalent in net removals by 2030. Monitoring relies on geographically explicit accounting approaches to ensure transparency and accountability across land categories such as afforested and deforested land, cropland, wetlands, and harvested wood products.

Given the breadth and intersectoral nature of land-use impacts, SEA is particularly relevant. SEA provides a framework to evaluate environmental trade-offs, align policies, plans, and programmes, enabling the integration of environmental considerations into strategic decision-making. By adopting a significance-based approach, SEA concentrates analytical attention on those environmental components most likely to be affected by proposed land-use configurations, such as biodiversity, soil functions, water systems, landscape structure, and climate-related risks. Rather than attempting exhaustive coverage of all possible impacts. In doing so, it strengthens the integration of environmental considerations into spatial planning processes and supports more informed, transparent, and ecologically responsive land-use strategies.

3.1.1. Land-use sustainability issues

Current land-use systems in Europe and globally face mounting sustainability pressures. Urban expansion and infrastructure development increase land take and soil sealing, reducing agricultural capacity and ecosystem functions. Intensive agriculture contributes to soil degradation, water stress, and biodiversity decline, while forest and wetland loss weakens carbon sequestration. Collectively, these trends undermine land's role as a net carbon sink and threaten long-term ecosystem resilience.

Looking ahead, competing land demands are expected to intensify. Climate mitigation measures such as afforestation and bioenergy crops, biodiversity targets, food security, and continued housing and infrastructure expansion all compete for limited space. Without strategic coordination, these objectives risk generating trade-offs and spatial conflicts that compromise sustainability outcomes.

Strategic Environmental Assessment (SEA) addresses these challenges by evaluating land use impacts across governance scales. At the national level, land use shapes climate mitigation, biodiversity protection, and food security through carbon accounting and ecosystem management strategies. At the regional level, spatial planning influences cumulative and cross-boundary effects, including habitat fragmentation, transport corridors, and peri-urban expansion. At the municipal level, land-use decisions directly determine urban form, emissions patterns, access to green infrastructure, and social equity outcomes.

Future trajectories therefore depend on integrated, multi-level planning. Strengthening carbon sinks, preventing irreversible land degradation, and reconciling development with ecosystem integrity require anticipatory spatial strategies rather than reactive responses. In this context, SEA serves as a preventive instrument, guiding land-use decisions toward climate neutrality, biodiversity protection, and resilient territorial development.

3.1.2. SWOT and crossed-SWOT analyses for the land-use sector

In order to go in depth into the sector characteristics, a SWOT analysis, showing pros and cons of land-use, has been carried out. The matrix is the fruit of a process of joining and overlapping of several others SWOTs, which has been realised starting from a former general brainstorming: once the main aspects of the sector were pointed out in a general scheme, those gathered in six different analyses, which were further condensed in a unique overarching matrix (table. 1) showcasing the ultimate definition of strengths, weaknesses, opportunities and threats.

In particular, what emerges from this first matrix is that the land use sector, on the one hand, presents non indifferent points of strengths. It, indeed, thanks to the vast amount of forests and wetlands that it comprehends, functions as a carbon sink, providing benefits in terms of ecosystem services and regulation. In a similar way, it also supplies economic benefits, this time by exploiting other subsectors, such as agriculture, industry and economic uses in general. Besides carbon sequestration and economic advantages, land use is also fundamental for its capacity of making food and growable resources, essential for the fulfilment of human needs. Moreover, as introduced before, land use can contribute to the creation of employment opportunities, directly linked to community well being, and

all kinds of ecosystem services. It represents the society's fundament, organising transformation in an overarching scale and regulating the management of capital, such as greenery and extractions. In addition to all these specific characteristics, the sector also acts as a loom for all kinds of transformations on different scales and, synchronising different sectors, for the reaching of cross-sectoral sustainability goals.

On the other hand, however, land use foundation is soil, which is a finite resource and, for this reason, its fertility and quality is often overlooked, giving priority to more lucrative transformation. However, construction and building use is not the only problem linked to the scarcity of land. Even when the soil is used for producing food it could arise problems linked to its usage and exploitation: high pollution and excessive water use from agriculture can damage the soil irreparably. Those human actions negatively affect the source in two ways: they pollute the soil, damaging biodiversity, but also causes habitat fragmentation. Overexploitation and mismanagement also causes an increment of Greenhouse gasses. Besides the environmental problems, there are also important political aspects that must be taken into account. One of those is represented by conflicts of interests between uses, since often one use excludes the others, and policy incoherence, which gives more importance to most profitable transformations, not considering their negative aspects. It is also common that general needs get in conflict with individual ones, which often prevails, entailing inequalities and treatment disparities. The last weakness linked to soil and land use is perhaps the most important: it concerns the irreversibility of transformations, which makes this sector a very delicate system, where every choice must be planned, analysing every single consequence and externality.

Besides strengths and weaknesses, the SWOT analysis also shows external factors. Those, as for the internal ones, are divided in two categories. On the one hand, there are opportunities, which, for the studied sector, are the possibility of new findings and technologies, which in future can ensure a better management of land use. Moreover, a better policy ground can promote more sustainable choices, also as regards financial aspects and global targets. Opportunities can be seen also from the ecological point of view: ecosystem restoration and landscape protection can bring benefits under several perspectives. This entire frame must be supported by a coordinate inclusive cross-scale governance which, integrated with a proper territorial governance, can ensure the social, ecological and economical sustainability of the sector.

However, opportunities are balanced by threats which, on the other hand, can concern the evolving situation of climate change and global warming, with always more frequent catastrophic events; the uncontrolled urbanisation, whose effects can be similar to the previous ones; and the loss of biodiversity, which affect habitats and ecosystem services. Moreover, economic interests and instability, an already compromised situation and conflicts between stakeholders menace the stability of the sector, making interventions complicated and delicate. To these problems is added the population growth and the pressure on the job market provided by the nowadays technological shift.

STRENGTHS	WEAKNESSES
S1. Functions as a major carbon sink (forests, wetlands)	W1. Finite land availability and declining soil quality under development pressure
S2. Provides economic benefits through agriculture, forestry, industry, and trade	W2. High pollution and excessive water use from agriculture and industry
S3. Core source of food production and growable resources	W3. Negative biodiversity impacts from intensive land use and fragmentation
S4. Supports employment, livelihoods, and community wellbeing	W4. Overexploitation and mismanagement
S5. Provides ecosystem services	W5. GHG emissions from land-use activities(land conversion, soil disturbance, and agricultural practices)
S6. Foundation for societies, infrastructure, culture, and heritage	W6. Conflicting land demands across sectors
S7. Strong natural capital and greenery	W7. Fragmented governance and policy incoherence
S8. Enables cross-sectoral sustainability goals (SDGs, energy, transport)	W8. Social and environmental justice gaps (general needs overlooked)
S9. Can coordinate transformation across scales via planning	W9. Least profitable sectors risk neglect
	W10. Irreversible land-use decisions with long-term impacts
OPPORTUNITIES	THREATS
O1. Research and development enabling climate-smart forestry, regenerative agriculture, smart cities	T1. Climate change impacts: floods, droughts, wildfires, heat stress
O2. Sustainable land-use planning integrating housing, mobility, and green infrastructure	T2. Escalating competition between urban expansion, food production, bioenergy, and conservation
O3. Policy innovation prioritizing ecological value over short-term profit	T3. Biodiversity loss and ecosystem degradation
O4. Sustainability-oriented financing (carbon removals, climate contracts)	T4. Economic instability if ecosystems stop providing services
O5. Net carbon removal targets transforming land into a climate solution	T5. Planetary boundaries already under pressure
O6. Ecosystem restoration and biodiversity enhancement	T6. Conflicts between stakeholders and governance levels
O7. Rising valuation of cultural landscapes for tourism and rural economic diversification	T7. Rising land prices, limiting access for farmers and housing
O8. Inclusive, cross-scale governance and coordination	T8. Population growth increasing pressure on land and resources
O9. Efficient zoning and land division to optimize land	T9. Job displacement due to technological shifts
O10. Potential for sustainable resource extraction and management	

Table 1: SWOT Analysis

As a further step, the cross SWOT has been realised by relating the four factors to each other. As a result, it emerges how to leverage the carbon sink capacity of forests and wetlands to meet climate neutrality targets and access sustainability-oriented financing. Use economic returns from agriculture, forestry, and trade to fund alternative farming techniques and climate-smart investments that increase productivity without expanding land use. Strengthen ecosystem restoration and tourism to generate revenue while enhancing biodiversity and cultural identity. Apply research, innovation, and advanced planning tools to improve food production efficiency and land management. Foster cross-sectoral collaboration, linking land use with energy, transport, and urban planning, to create integrated and resilient territorial systems.

Moreover, utilize wetlands and forests as natural barriers to reduce flooding, heat islands, and other climate-related risks. Protect high-quality agricultural land through strict zoning to safeguard food security under population growth pressures. Apply strategic land-use planning to limit damage from competing land demands and rising land prices. Use ecosystem services to counteract environmental degradation and reduce climate vulnerability. Support workforce transition policies that rely on the sector's economic importance to minimize job displacement during sustainability shifts.

On the other hand, use research funding and innovation to improve soil health, water efficiency, and biodiversity management. Rethink land allocation policies by prioritizing ecological value over short-term economic incentives. Strengthen governance integration and participatory planning frameworks to reduce conflicts of interest. Promote circular and local resource systems to reduce pollution and overexploitation of finite resources. Adopt alternative agricultural practices to decrease emissions and excessive water use while increasing long-term productivity.

Implement strict safeguards and zoning regulations to prevent irreversible land degradation and urban sprawl on vulnerable land. Reduce emission-intensive and pollution-heavy practices, particularly in agriculture and industry. Regulate resource extraction to protect soil health and ecosystem stability. Reuse and redevelop existing built environments to limit additional land taken under urbanization pressure. Align policies across governance levels to avoid fragmented decision-making and prevent business-as-usual trajectories that accelerate environmental and socio-economic decline.

	OPPORTUNITIES	THREATS
S T R E N G T H S	S1.O5. Leverage carbon sink capacity of forests and wetlands to meet climate neutrality targets, access Climate Neutrality Contracts, and secure sustainability-oriented financing.	S1.T1. Use forests and wetlands as natural barriers to mitigate urban flooding, heat islands, and climate-related risks.
	S2.O1. Use economic benefits from agriculture, forestry, and trade to invest in climate-smart and alternative farming techniques that increase productivity without expanding land use.	S2.T4. Protect growable resource areas through land-use planning to ensure food security under population growth and planetary boundary pressures.
	S4.O2. Apply research, innovation, and advanced planning tools to improve land-use efficiency, food security, and long-term resilience.	S4.T9. Support workforce stability and wellbeing by using the sector's economic importance to manage transitions and reduce job displacement risks.
	S6.O7. Strengthen socio-economic development and wellbeing through sustainable production, ecosystem restoration, and tourism linked to cultural and natural heritage.	S5.T3. Rely on ecosystem services and natural capital to reduce vulnerability to climate change impacts and environmental degradation.
	S9.O8. Utilize the multidimensional nature of land use to foster cross-sectoral collaboration (energy, transport, urban planning) and create integrated sustainability solutions.	S10.T2. Employ land-use planning and zoning to manage urbanization, competing land demands, and rising land prices.
W E A K N E S S E S	O1.W1. Use research, innovation, and funding mechanisms to improve soil health, water efficiency, biodiversity, and land management.	W1.T2. Implement strict zoning and safeguards to prevent urbanization on vulnerable, low-fertility, or culturally significant land.
	O2.W3. Rethink land allocation policies to prioritize ecological value and nature-based solutions over short-term economic incentives.	W2.T1. Reduce emission and pollution-intensive practices, particularly in agriculture and industry, to avoid exacerbating climate change and ecosystem degradation.
	O3.T6. Strengthen governance, policy integration, and participatory planning frameworks to reduce conflicts of interest and fragmentation.	W4.T4. Regulate resource extraction and land management to prevent irreversible soil degradation and biodiversity loss.
	O4.T7. Use sustainability-oriented financing to address land scarcity, improve efficiency, and support equitable development.	W7.T6. Align local, regional, and supralocal land policies to avoid business as usual trajectories leading to environmental and socio-economic collapse.
	O8.T5. Promote circular and local resource systems to reduce pollution, excessive water use, and pressure on finite resources.	W8.T8. Plan proactively for population growth and urban expansion by reusing the built environment instead of increasing land take.

Table 2: Cross-SWOT Analysis

3.2 The Linköping Masterplan

3.2.1 Description of the plan

The Linköpings Masterplan (2010), formally adopted as the Översiktsplan för staden Linköping, constitutes the municipality's statutory comprehensive plan and establishes the long-term spatial structure for the entire municipal

territory. It sets out strategic directions for the allocation and management of land and water areas and outlines the anticipated development of the built environment. As such, it provides the overarching framework guiding subsequent detailed development plans, infrastructure investments, and permitting decisions.

The plan defines where and how growth should occur, identifying priority development areas for housing and employment, regeneration within the existing urban fabric, and protection zones for green structure, agriculture, and cultural landscapes. A key spatial strategy is to concentrate new construction within or adjacent to the existing city and along the proposed high-capacity public transport corridor (LinkLink), rather than allowing outward expansion. This corridor-based structure links new neighbourhoods, services, and workplaces to rapid public transport, walking, and cycling networks, while redesigning major roads to reduce through-traffic in central areas.

The scope therefore spans land use, housing provision, transport systems, green and blue infrastructure, environmental protection, and cultural heritage, integrating these sectors within one coordinated framework. In terms of time frame, the masterplan is not limited to a short programme period but establishes a long-term strategic direction for several decades, guiding development progressively as the city grows. Adopted by the City Council in 2010, it functions as an enduring reference for future planning and investment decisions rather than a fixed implementation schedule.

3.2.2 The importance of planning

The 2010 Linköping Master Plan (Översiktsplan för staden) is needed to manage rapid population growth from 165,000 residents in 2022 (*Eurocities, 2022*) toward the predicted 170,000 target, preparing for up to 200,000 through proactive land-use strategies. Uncontrolled expansion would erode the city's 700-year cultural heritage and key ecological assets like the Stångå river corridor and Tinnerö oak landscape.

Sweden's Environmental Code mandates Strategic Environmental Assessment (SEA) for such municipal plans (*European Commission, n.d.*), ensuring early integration of environmental, cultural, and social factors into densification decisions. This addresses land scarcity by prioritizing infill development within existing urban areas, avoiding sprawl that increases infrastructure costs and fragments green spaces.

The plan fosters a compact, cohesive city (*Linköping kommun, 2010*). form that preserves heritage as an "identity-builder" while enhancing green infrastructure for biodiversity and recreation. Outcomes include reduced car dependency, lower emissions, cost-efficient resource use, and sustained livability amid growth pressures.

This report critically assesses this goal by evaluating densification alternatives against environmental baselines, identifying potential trade-offs in heritage preservation and ecosystem services, and recommending mitigation measures to ensure land-use strategies align with sustainability objectives (*European Commission, n.d.*). Building on this analysis, it strengthens plan implementation through monitored indicators for green space retention and cultural integration, enabling adaptive management for long-term urban vibrancy.

3.3 SEA report components

3.3.1 Strategic Reference Framework (SRF)

The Strategic Reference Framework (SRF) structures the application of SEA to the land-use dimension of the Master Plan (Översiktsplan) of Linköping. Rather than evaluating the municipality as a whole, the SRF specifically examines how land allocation, spatial structuring, and development strategies within the Master Plan influence long-term environmental sustainability.

The framework is grounded in a Strategic Thinking SEA (ST4SEA) approach. This means the assessment does not just identify environmental impacts of predefined proposals; instead, it evaluates and guides land-use development trajectories at the strategic level.

3.3.1.1. SEA Process

The primary aim of the SEA was to determine whether the comprehensive plan's long-term land-use strategy aligns with environmental sustainability objectives and to identify which spatial configurations best support ecologically resilient urban development. Aim

To achieve this, the SEA evaluates and guides the land-use strategy by

- Identifying key sustainability challenges
- Evaluating long-term and cumulative environmental effects
- Developing and comparing alternative spatial strategies
- Supporting informed and environmentally responsible decision-making

The SEA process was structured as an iterative analytical sequence aligned with Strategic Thinking SEA principles.

1. Review of the Master Plan (Land-Use focus): The land-use structure of the Master Plan (Linköping kommun, 2010) was examined to identify key development priorities and spatial patterns.
2. SWOT Analysis: A SWOT analysis was conducted for the land-use sector to identify strengths, weaknesses, opportunities, and threats influencing sustainable spatial development.
3. Cross-SWOT Analysis: The SWOT findings were synthesised to identify strategic leverage points and risk patterns, shifting from descriptive diagnosis to strategic interpretation.
4. Formulation of Strategic Objective: An overarching sustainability objective for the land-use sector was formulated to guide subsequent assessment stages.
5. Identification of Critical Decision Factors (CDFs): Key thematic areas where land-use decisions generate significant environmental implications were identified.
6. Development of Assessment Criteria and Indicators: Assessment criteria were formulated for each CDF, followed by measurable indicators enabling structured evaluation of land-use performance.
7. Development of Strategic Options: Alternative land-use development trajectories were formulated and assessed against sustainability criteria.

8. Risk–Opportunity Analysis: A risk–opportunity matrix was applied to evaluate trade-offs, cumulative effects, and structural vulnerabilities for the strategic options to generate the recommended strategic options..
9. Monitoring Framework: Monitoring indicators were identified to support long-term follow-up and adaptive planning.

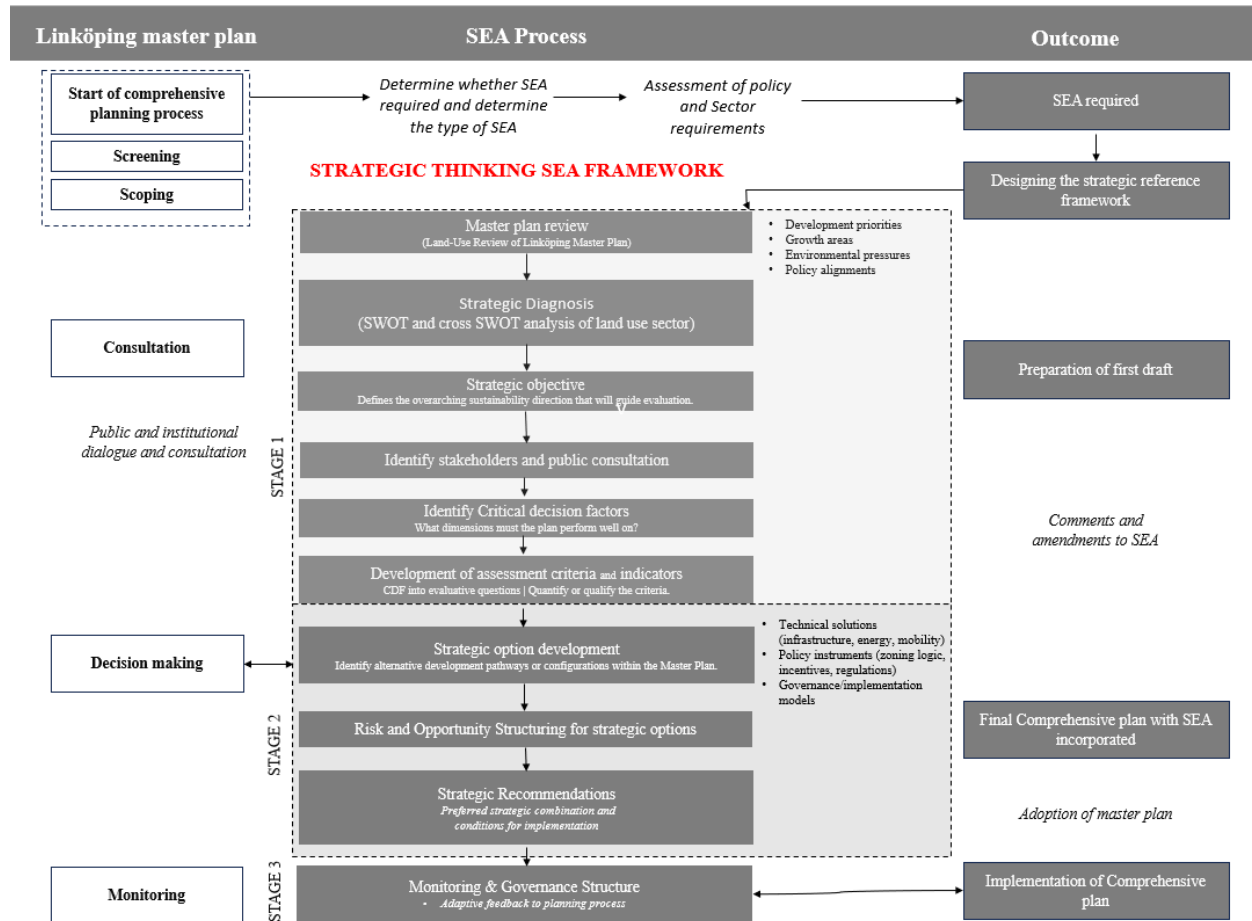


Figure 6: Flow diagram showing Intended analytical framework guiding the SEA of the Comprehensive Plan of Linköping. Source: Authors, adapted from (Azcarate, 2026)

3.3.1.2. Scope

Land-use allocations and spatial strategies within the Master Plan (Linköping kommun, 2010), including urban development areas, agricultural land, ecological systems, infrastructure corridors, and cultural landscapes. The spatial scope of the work embraces the municipal territory as defined in the Master Plan (Figure 7); as regards the

temporal span, it consists of a long-term strategic planning horizon of the comprehensive plan, i.e 2030.



Figure 7: Scope of master plan Source: Linköping kommun. (2010)

3.3.1.3 Relevant International and National Legislation and Policy Framework

As outlined in Section 1.2.3, the SEA application is guided by relevant policy and legal frameworks to ensure compliance and alignment with sustainable land-use objectives.

At the European level, the SEA is conducted in accordance with the Directive 2001/42/EC (European Commission., 2001). The Directive requires that plans and programmes likely to have significant environmental effects, particularly those setting the framework for future development consent, such as comprehensive land-use plans, undergo environmental assessment. It mandates early integration of environmental considerations, evaluation of reasonable alternatives, public consultation, and monitoring of significant effects. For Linköping's Master Plan, this Directive provides the overarching legal obligation to assess how strategic land-use decisions may influence environmental quality, resource use, and long-term sustainability.

At the national level, the SEA Directive is transposed into Swedish law through the Miljöbalken (1998:808). Chapter 6 of the Environmental Code regulates environmental assessment of plans and programmes and defines procedural requirements for screening, scoping, consultation, environmental reporting, and follow-up. Under this framework, municipalities are responsible for ensuring that comprehensive plans integrate environmental considerations and assess significant environmental effects prior to adoption.

Beyond legal compliance, the SEA of Linköping's Comprehensive Plan is informed by the 2030 Agenda for Sustainable Development. The sustainability objectives guiding this SEA are aligned with the Sustainable Development Goals explicitly referenced in the sector analysis, namely SDG 11 (Sustainable Cities and Communities), SDG 13 (Climate Action), and SDG 15 (Life on Land). For Linköping, these goals are relevant as the Comprehensive Plan's land-use decisions directly affect climate resilience, biodiversity, and sustainable urban growth.

The SEA is intended to be conducted in parallel with the drafting of the comprehensive plan (Översiktsplan), in accordance with the requirements of the Directive 2001/42/EC (*European Commission, 2001*) and the Swedish Miljöbalken (1998:808). By embedding the assessment within the strategic stage of municipal planning, the process ensured that environmental considerations informed the structuring principles of spatial development visible in Figure 6. The SEA is not intended as an external audit but as an internalised framework guiding the long-term transformation of Linköping towards a vibrant and environmentally sustainable urban form.

3.3.2 Governance framework and stakeholders

The successful implementation of the Linköping Masterplan, balancing urban density with environmental preservation, relies on a multi-level governance structure. The following analysis details the roles, power dynamics, and interests of the four primary stakeholder sectors as they relate to the SEA objectives. The goal is to create a collective vision and objectives that are shared and engaged by all stakeholders and a participative process (Partidário, 2012). Once all stakeholders had been identified, the next step consisted of assessing their relevance to the study. To this end, their interest (the relationship between the stakeholder and the study area), power (the capacity to influence the state of the art), and importance (their relevance to the proposed actions) were evaluated. Each indicator was assigned a score based on the normalization of a qualitative nine-level scale ranging from “very low” to “very high”. Figure 8 illustrates their distribution on a Cartesian plane. In the upper-left quadrant are the actors who must be kept satisfied, namely housing companies, landlords, and the Swedish State. On the right-hand side of the chart are the most relevant stakeholders, those directly involved in managing the actions. The most important among them are the municipality and the military, which show the highest level of interest in the area. In the lower part of the diagram are the actors who must be monitored and those who must be kept informed. The former include the Swedish Agency for Water and the Swedish Heritage Protection Agency, while the latter consist of those living in the area, such as farmers, residents, and local economic activities.

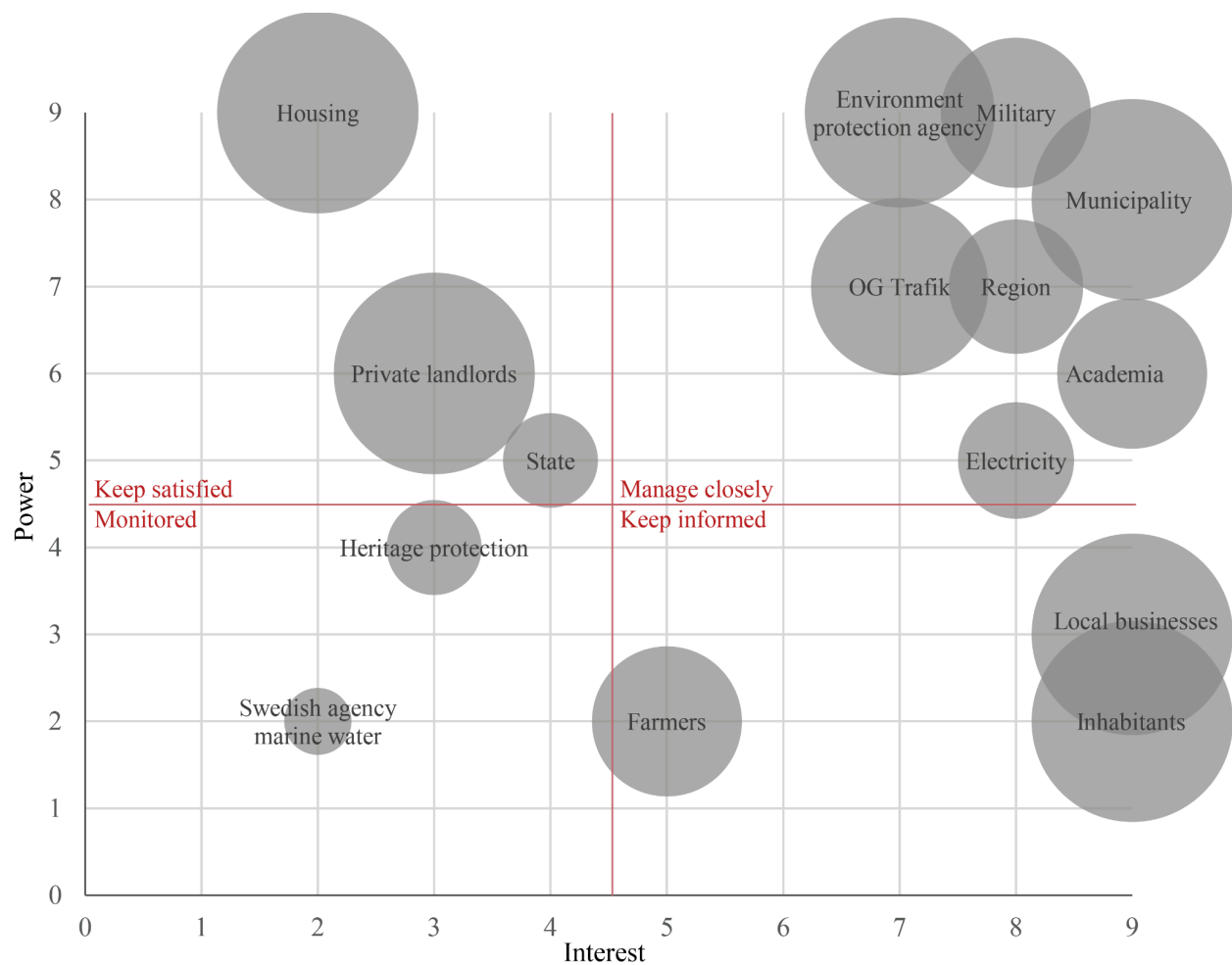


Figure 8: Stakeholders and main actors

Stakeholder	Interest	Power	Importance	Stakeholder	Interest	Power	Importance
Housing	2	9	9	Academia	9	6	5
Private landlords	3	6	9	Region	8	7	4
State	4	5	2	Municipality	10	9	9
Heritage protection	3	4	2	Military	8	10	5
Swedish agency water	2	2	1	Electricity	8	5	3
Farmers	5	2	5	Local activities	9	3	9
OG Trafik	7	7	7	Inhabitants	9	2	10
Environment protection	7	9	8				
Quali-quantitative conversion scale		Very low	Low	Moderate	High	Very high	
		1	3	5	7	9	

Table 3: Ranking rubrics

3.3.2.1 Public sector

Linköping Municipality (Linköpings kommun) serves as the primary administrative authority responsible for the development and implementation of the Masterplan (Översiktsplan), which is currently under analysis (Linköpings kommun, 2010). The plan was authored by the Environmental and Urban Planning Department (Miljö- och samhällsbyggnadsförvaltningen) and legally adopted by the Municipal Council (Kommunfullmäktige). As the entity holding the planning monopoly in Sweden, the municipality possesses high power to dictate land-use patterns and infrastructure priorities, such as the "LinkLink" high-efficiency transit system. Their high interest is driven by the mandate to manage the city's significant growth, projected to reach up to 200,000 inhabitants, while balancing this with environmental sustainability and national interests (Riksstyrelsen). Additionally, the County Administrative Board (Länsstyrelsen) acts as a key public stakeholder by providing oversight on regional water management in the Stångån river and preserving cultural heritage sites.

3.3.2.2 Private sector

The private sector, encompassing real estate developers, industrial leaders like SAAB, and the technology firms within Mjärdevi Science Park, holds high power in the governance framework because the physical execution of the Masterplan relies on private investment and construction capacity. This sector's high interest is rooted in the economic implications of urban planning; for instance, the expansion of the inner city and the development of the "Ostlänken" high-speed railway directly influence property values and business logistics (Trafikverket, 2025). Collaboration between the public and private sectors is exemplified by the formation of FlygMex, a development company established to expand Linköping City Airport and its adjacent business parks. These actors are essential for transforming the plan's vision of a "dense and green" city into functional residential and commercial spaces.

3.3.2.3 Civil society

Civil society, which includes local residents, environmental NGOs, and various interest groups, is characterized by high interest but low power regarding direct legislative decision-making. The Masterplan acknowledges the "invaluable" contributions of engaged citizens whose feedback during the consultation (samråd) and exhibition phases led to critical revisions of the plan. These stakeholders are primarily concerned with "livability" factors, such as the preservation of the Tinnerö oak woodland (Eklandskap) and the enhancement of recreational corridors along the Stångån river (Länsstyrelsen Östergötland, n.d.). While they do not hold planning authority, their influence is exerted through democratic participation and the right to appeal specific development projects that may threaten local biodiversity or the availability of urban green spaces.

3.3.2.4 Academia

Academia, primarily represented by Linköping University (LiU), occupies a role with medium power and medium interest within the city's governance structure. The Masterplan explicitly designates LiU as a "world-class university" and reserves specific "UN-areas" for its future physical expansion (Linköpings kommun, 2010), highlighting its importance to the city's identity as a knowledge hub. The university's interest lies in ensuring the city remains attractive for students and researchers through sustainable transport links and adequate housing. While

academia does not dictate policy, its power stems from the "Triple Helix" model of collaboration, where the university provides the research, data, and innovation necessary to implement complex sustainability measures, such as the city's technical energy supply and waste management systems (Linköping University, 2010).

3.3.3 Results from Interviews

To gain insight into residents' perspectives on land-use and urban development in Linköpings municipality, two former urban planning students from Linköpings University, Anna (23) and Linnea (24), were interviewed. Both interviewees relocated to Linköping to study urban and regional planning for three years. The objective of conducting these interviews was to understand how students perceive the city's spatial structure, sustainability, and livability, as well as finding opportunities and challenges for urban development (section 3.3.3.1 to 3.3.3.4). Another interview was conducted via mail with Mikael Sonesson, a comprehensive planner at Linköpings municipality. Mikael has worked for the municipality for six years. Furthermore, he lives in Linköping, approximately 13 kilometers from the urban center. The objective of the interview was to gain insight into the current state of municipal land-use strategy. Additionally, the interview was conducted to track key challenges and planning priorities within the framework of a Strategic Environmental Assessment (Section 3.3.3.5 to 3.3.3.9).

3.3.3.1 Living and affordability

The respondents agreed that Linköping is an attractive university city. The city has a strong student life with excellent educational programs. The respondent recommends other students to live in Linköping. Mainly for its good programs, accessibility, bike culture, and closeness to nature. However, one respondent noted that while Linköping is suitable for temporary study, long term residency may be more appealing in a larger city. It was agreed that the housing situation in Linköping was challenged by its high prices, although corridor rooms are more affordable. Smaller apartments suitable for students are expensive. Limited housing is not the concern, new housing developments have been built. However, despite the ongoing new housing developments, affordability remains a key concern for students. The university in linköping was described to be separated from the city, resulting in what was described as "two separate worlds". This spatial division can create social separation. This separation could lead to university students feeling isolated from broader city life.

3.3.3.2 Transportation and job

Linköping is a technology oriented city, with a strong focus on innovation. Companies such as SAAB are located in Linköping. Although, compared to Stockholm, job opportunities are more limited, the engineering and technology sector may be strong in Linköping, it is described that inhabitants commute to Stockholm for work. The city growth pattern has been described as banana shaped expansion, leading to spatial fragmentation. Respondents emphasized the importance of improving connectivity between outer residential areas and the city center to support linköpings development. This aligns with the municipality's overview plan, reshaping the banana shape into a connected circle.

Cycling is the main form of transportation. Due to the municipality's well maintained and heated up bike lanes, biking is accessible all year around. Furthermore, public transportation is affordable and cheaper during off peak hours. Public transportation is considered efficient for travel to the city center. However, cross city bus connections, particularly between outer residential areas, were seen as less accessible and should be improved. This highlights that mobility planning is crucial for good access across the urban area.

3.3.3.3 Environment

Linköping is considered a green and environmentally conscious city. Nature reserves, such as Tinnerö Eklanskap, are easily accessible. The city integrates parks and forests into urban areas. Sustainable efforts include resource efficient new development, green energy initiatives, and strong infrastructure. However, access to water for recreation was described as limited, as lake Roxen for example, is not easily accessible through public transport due to poor connectivity.

Lastly, the city is perceived as vibrant and safe, especially in central areas. However, certain neighbourhoods, such as Skäggetorp and Berga, were perceived as less safe due to reported violence. Noise conflicts between students and families were noted in mixed residential areas such as Vallastaden. These social challenges suggest the importance of integrating good planning to balance diverse urban populations and prevent social isolation.

3.3.3.4 Development and restrictions

The respondents highlighted collaborative planning efforts between Linköping and Norrköping, with the ambition to strengthen the region. Projects such as Vallastaden and the proposed high speed rail project Ostlänken were discussed. Ostlänken was seen as significant for the future development, although concerns were raised about the placement of the new station and accessibility. This puts a focus on the importance of using the land sector of the municipality to improve transport infrastructure in shaping urban growth and regional connectivity.

However, military land restrictions were identified as a structural constraint in urban expansion, as the military holds veto power over certain development areas. Such constraints put a need for strategic planning to optimize urban growth, while, at the same time, respecting military boundaries. Lastly, students find Linköping to be an innovative, technology oriented, and environmentally conscious city. Its strong biking culture and regional collaboration makes the city sustainable and environmentally conscious. Challenges that Linköping faces include housing affordability, social separation between students and residents, and spatial fragmentation. Furthermore, limited recreational access to water needs to be improved. Addressing these issues through good integrated planning. Improved connectivity is crucial to support sustainable urban development.

3.3.3.5 Identified challenges

According to the interview, key challenges were identified, These can be considered major challenges within the land-use sector. These challenges include management of agricultural land, climate change adaptation, the development of the Ostlänken railway, and segregation. Agricultural land was a recurring topic, as a source of

conflict. Linköping tries not to take agricultural land. However, it was stated that it sometimes is necessary. Additionally, improving accessibility for different transport modes and reducing the share of car travel were in existing tension. These conflicts showcase the difficulty to balance environmental, infrastructural, and social objectives. Moreover, the alignment of the Ostlänken railway was described as an opportunity, as it creates new development possibilities. However, it could be a challenge as it simultaneously affects spatial properties.

3.3.3.6 Current framework

Linköping operates under several environmental protection programs, including the nature conservation program, the climate energy program, the climate adaptation program, the chemical program, the action program for noise and air quality, and the water service plan. These programs were brought during the interview through mail. The purpose of these programs is to collectively guide environmental governance within the municipality.

3.3.3.7 Sustainability

The municipality has developed specific guidelines for ecosystem services and ecological compensation. Open stormwater solutions are mentioned in the interview to promote a nature-based approach. The open stormwater would simultaneously manage water and enhance ecosystem services. Contrary to Linköping's own guidelines, the municipality states that it tries to align its planning with the global sustainability goals (Sustainable Development Goals, SDGs). However, the interviewee writes that these are difficult to operationalize in practical planning. Instead, Swedish national environmental objectives are considered more directly applicable to municipal activities.

3.3.3.8 Growth and direction

A central aim, planned for the long term, is to create a compact urban shape. The purpose is to accommodate a projected population of 250,000 inhabitants. The vision would benefit from the implementation of a strategy that concentrates on development around urban areas, rather than allowing growth in rural direction. However, tension between the usage of agricultural land and urban expansion is active. As previously mentioned, the interviewee highlighted that the municipality attempts to avoid developing agricultural land, it is sometimes necessary to enable urban growth. Additionally, the municipal plan is to concentrate development around selected areas while limiting scattered development in rural areas. This underlines that the municipality has a planning approach that aims to balance urban compaction with agricultural land maintenance.

3.3.3.9 Stakeholders

Key stakeholders, in relation to land-use planning, include the business sector, citizen, civil society, state authorities, and internal municipal organizations. According to the interviewee, conflicts of interest are balanced through professional assessment, political direction, and in accordance with municipal policy documents.

3.3.4 Problem statements and SEA elements: from CDFs to indicators

Formulating the problem statement, in which the general direction of the assessment is defined and formulated as a strategic objective, is one of the most important steps in the SEA procedure. In this SEA, the strategic objective is defined as follows: "Making Linköping a vibrant and attractive city, while respecting nature and the historical heritage of the town." This strategic objective points out the main conflict in land-use planning in Linköping, which is how to develop the city in terms of growth, innovation, and attractiveness without being a burden to nature and the identity of the town.

Typical characteristics for Linköping are its urban development, economy, which is linked to Linköping University, and the presence of high-technology industries. These factors exert pressure on land, infrastructure, and natural resources. However, part of the identity of the municipality is its agricultural land, green corridors, water resources, and historically significant town center. Thus, a dilemma can be observed: development is required, but the character and location of the development will determine whether there is degradation of nature and loss of identity or whether development can help to strengthen nature and ensure the continuity of identity.

The Critical Decision Factors (CDF) were established directly from the strategic objective. They define the ambition of vibrancy and attractiveness to the areas of decision-making. CDFs define the areas of leverage where land-use decisions will have the most effect on whether the development path of Linköping will enhance or degrade the environment, cultural heritage, and social vitality.

Strategic objectives

Making Linköping a vibrant, and attractive city, respecting nature and the historical heritage of the town.

CDFs	CDF Definition	Targets	Assessment Criteria	Indicators
H. Cultural heritage	Managing, preserving, and integrating tangible (monuments, ruins, historic buildings) and intangible (traditions, indigenous practices) elements into land use planning.	H1. Preserve historical buildings	Ac.H1. Reduction of demolitions of buildings in the old town	i) Percentage of old buildings demolished in the town center ii) Amount of buildings that are culturally protected (K-märkta)
		H2. Increase attractiveness for cultural tourism	Ac.H2. Promotion of informative, interactive engagement programs associated with the heritage sites	i) Annual visitor numbers to heritage districts ii) Number of cultural sites integrated into public space strategies
I. Industry renovation	Modernizing or upgrading an industry to improve its efficiency, competitiveness, technology, and overall performance.	I1. Enhance the green industrial sector aligning it with the SDGs and the carbon-neutral goals of Sweden	Ac.I1. Production of carbon emissions from the industrial sector	i) CO ₂ emissions per hectare of industrial land ii) Percentage of renewable energy used

U. University and scientific development	Improving and expanding the university's infrastructure, research capacity, student services, and overall institutional quality.	U1. Make Linköping University campus carbon neutral	Ac.U1. Production of carbon emissions from the university	i) Quantitative level of percentage fossil fuel used
		U2. Attract students and talents	Ac.U2. Number of new student accommodation	i) Square meter of new student housing
E. Ecosystems and urban ecology	systematic integration of natural, green, and blue infrastructure, along with biological processes in land use planning.	E1. Integrate more greenery in the urban fabric and reconstruct the dialog between the city and the river	Ac.E1.1. Availability of new trees and green areas Ac.E1.2. Accessibility and usability of the river	i) Distance (square meters) between residential buildings and trees/green areas ii) Quality of connection between residential areas and river
		E2. Reduce soil consumption	Ac.E2. Expansion of neighbourhoods outside of the built area	i) Amount of square meters of unspoiled soil consumed
		S1. Create safe, inclusive and diverse public areas	Ac.S1.1. Quality and safety of spaces Ac.S1.2. Availability of diverse functions and uses	i) Qualitative data about safety perception (Survey-based safety perception index) ii) Percentage of diverse uses and functions
S. Civil society and vibrant city	To create a lively, dynamic urban area characterized by active economic activity, cultural diversity, social engagement, innovation, and a high quality of life.			i) Data about quality of existing bicycle paths
		S2. Sustainable public transport and mobility	Ac.S2. Availability of active mobility infrastructures and possibility to move using public services	ii) Distance (square meters) between residential buildings and transport hubs/stations iii) Travel time between residential areas and work places
		S3. Improve the business centre, also attracting and enhancing entrepreneurship and business activities	Ac.S3. Increase policy to enhance economic activity	i) Percentage of employees and working inhabitants that works in the city
		S4. Foster the offer for more affordable and accessible housing	Ac.S4. Home availability and affordability	i) Amount of square meters of new or repurposed accommodation units ii) Policy price caps on rent for students and low-income population

Table 4: CDFs and assessment criteria

3.3.5 Assessment criteria and indicators

To further develop the CDFs, the assessment criteria and indicators will be disclosed. The identified CDFs are integrated themes aimed to focus the strategic decisions of the SEA (Partidario, 2012). To ensure a rigorous analysis and strategic thinking within the SEA, assessment criteria will be utilized in order to set a scope for the CDFs and correlated targets. The assessment criteria creates the assessment framework in combination with interrelated indicators for each assessment criteria to ensure thorough measurement of the sustainable change and trajectory (Partidario, 2012). Thus, the assessment criteria is utilized to provide further clarity and priority of what the CDFs entail and ensure alignment with project objectives and performance. Nonetheless, this is further assessed by the indicators. The role of the indicators is to function as an objective and specific indication of development, both quantitative and qualitative, of the SEA and strategic thinking (Partidario, 2012). Thereby, the assessment criteria and indicators, as a central part of the SEA for Linköping municipality, will be further explained and justified in the following section.

3.3.5.1. H: Cultural Heritage

H1: Preserve old buildings

The first identified CDF is cultural heritage. With this CDF the first target aims to ensure the preservation of architectural and cultural identity of the city. This thereby generates the target H1: Preserve old buildings. To ensure the trajectory of this target an assessment criteria has been constructed. Namely, reduce the demolition of old buildings in the old town (assessment criteria Ac.H1). This aims to ensure the preservation of the cultural heritage and cultural identity within the city through old buildings and urban form. To ensure the development aligns with the strategic objectives and targets, two indicators will be used to follow the trajectory. These two indicators will aim to answer the two following questions; (i) How many buildings were demolished in the town centre, and (ii) How many buildings have cultural protection (K-märkta) in the city.

The first indicator aims to quantitatively ensure that the currently standing old buildings are preserved and maintained. This will be measured by the percentage of demolished buildings in the old-town and the city centre, specifically old buildings and central cultural buildings. The central objective of this indicator is to ensure that the preservation of old and cultural buildings is maintained in order to uphold the cultural heritage of the city against other market and modernization forces (Vulpiani et al., 2026).

The second indicator aims to assess and ensure the preservation and institutional consideration of the culturally protected buildings (K-märkta) in the city (Länsstyrelsen Östergötland, n.d.). Thus, to apply a mixed-method assessment of the number of culturally protected buildings in the city as well as the maintenance of these buildings can ensure a holistic approach to the CDF cultural heritage. Additionally, to enhance the cultural identity of the city a qualitative indicator can be utilized in order to identify further buildings of central value and cultural heritage, of the citizens, within the city to broaden the culturally protected buildings coverage, to create a stronger protection of the overall cultural identity of the city.

H2: Increase attractiveness for cultural tourism

The second target for the cultural heritage CDF is to increase the attractiveness of cultural tourism in Linköping. This target will be assessed by promoting informative, interactive engagement programs associated with the heritage sites (assessment criteria Ac.H2.). It broadens the scope of the cultural heritage and includes further cultural heritage sites, such as parks and natural environments in addition to cultural buildings. This can entail information boards highlighting the history of core places in the city as well as informative tours and talks on intangible elements such as traditions and practices.

The assessment criteria will be evaluated using the indicator (i) concerning the annual visitor numbers to heritage districts. Thus, by applying a quantitative approach to assess the tourist attractiveness of cultural sites in the city can be estimated and tracked. This can be utilized in order to identify sites of importance for cultural tourism as well as sites which might need improvement. This indicator will be combined with (ii) analyzing how many cultural sites can be integrated into public space strategies. Namely, how cultural heritage can be considered and incorporated into further urban planning to ensure long-term and sustainable cultural tourism in the city. Additionally, simple surveys at central museums or cultural tourist sites can be implemented to evaluate the satisfaction of tourists after visits, which can help improve the quality of these services to ensure a long-term maintenance of this target.

3.3.5.2. I: Industry renovation

II: Enhance the green industrial sector

The second CDF aims to modernize and restructure the industry services within Linköping towards more sustainable and innovative solutions and practices. This is further specified through the target; Enhance the green industrial sector aligning it with the SDGs and the carbon-neutral goals of Sweden. This target will be assessed by the assessment criteria (Ac.II.) Production of carbon emission from the industrial sector.

The assessment criteria will be evaluated by two indicators, (i) CO₂ emissions per hectare of industrial land, and (ii) percentage of renewable energy used. The first one will thereby follow the fossil fuel consumption of industries and thereby account for their carbon emission. This is of direct importance in relation to the target which aims to align Linköping's industries with the carbon-neutral goals of Sweden. The carbon-neutral goal aims to have a zero net emission of GHG gases by 2045 (Krisinformation, 2023). Thus, by ensuring little to no usage of fossil fuel within the industrial sector of Linköping, the municipality can align and contribute to these goals as well as provide a healthier and more sustainable environment for its citizens.

The second indicator focuses on the usage of renewable energy in a quantitative manner. Namely, by measuring the percentage of renewable energy used within the industries the municipality can track the enhancement of the green industrial sector. This indicator also aims to align the objectives with the SDGs, more specifically SDG 7, affordable and clean energy, SDG 9, industry, innovation and infrastructure, and SDG 12, responsible consumption and production (UN, n.d.). This indicator will thereby measure the green transition of the city and create further enhancement of green production and industry.

3.3.5.3. U: University and scientific development

U1: Make Linköping University campus carbon neutral

Furthermore, Linköping University is of central importance for the development of the city through its contribution to innovation and socio-economic values (LiU, n.d.). Nonetheless, with its central services and attractiveness for students, both national and international, this CDF aims to redesign the campus to make it more sustainable and carbon-neutral, to align with Sweden's environmental objectives. This will be indicated by the percentage of fossil fuel used at campus. Thus, this indicator aims to track the university's energy consumption, more specifically fossil fuel consumption. This entails heating and cooling of buildings, lights, and other technical services central to the University. Hence, by ensuring a sustainable campus the contribution by the university, at large, will provide inspiration, sustainability and incentives for further innovation and sustainable practices.

U2: Attract students and talent

Moreover, to further enhance the university and scientific development in Linköping, the second target is to attract students and talents. This is correlated to the assessment criteria (Ac.U2.) Number of new student accommodation. Namely, by enhancing the capacity of the university to receive students and provide accessible accommodation the assessment criteria will contribute to reaching the target, U2. Furthermore, the assessment criteria will be assessed by the indicator (i) square meters of new student housing, which directly quantifies the development trajectory of this target.

3.3.5.4. E: Ecosystems and urban ecology

E1: Integrate more greenery into the urban fabric

The fourth CDF is focused on the environmental aspects of Linköping. More specifically, the ecosystem and urban ecology in order to align with the strategic objective in terms of respecting nature and creating a vibrant and attractive city. The first target of this CDF is to integrate more greenery in the urban fabric and reconstruct the dialogue between the city and the river. To complement this target two assessment criterias will be implemented. These two assessment criteria are: (Ac.E1.1.) availability of new trees and green areas, and (Ac.E1.2.) accessibility and usability of the river. Firstly, the availability of trees and green areas is of central importance to the overall environmental quality of the city. This entails ecosystem and habitat protection, improvement of noise and air quality as well as social benefits through quality of public spaces and recreational areas. Secondly, the accessibility and usability of the river further enhances social and cultural capital through providing environmental areas for gathering and ensuring accessibility to these areas, which can improve aspects of environmental justice through disability adaptations.

The assessment criteria will be measured by the indicators; (i) distance (square meters) between residential buildings and trees/green areas, and (ii) Quality of connection between residential areas and rivers. The first indicator will utilize GIS in order to ensure residential areas proximity and accessibility to green areas. By using GIS, a buffer zone can be implemented between residential areas and green spaces to identify areas in need of more greenery. This

is important in order to ensure that the qualities and accessibility to green spaces is equal between residential areas and residents. The second indicator aims to ensure quality connections of residential areas, streets and pathways between the river and city. This entails safe passages and accessibility in terms of disabilities and for example prams. It focuses on measuring how accessible, safe, and engaging the riverfront is for residents, reflecting both physical and visual connections. By including the river in the ecosystem CDF, attention can be drawn to ensure the quality of river and river basin ecosystems in further planning. The overall importance of ecosystems in the city can improve ecological values, as well as improve climate adaptation and mitigation through ecosystem services.

E2: Reduce soil consumption

The second target within this CDF is to reduce soil consumption, in line with the central points brought forward by the interview with the municipal planner. This will be assessed through the assessment criteria (Ac.E2) expansion of neighborhoods outside of the built area. Thus, reducing consumption of undeveloped land to commercial or residential areas can preserve and protect ecosystems and biodiversity. By reducing soil consumption Linköping can develop without jeopardising the environmental qualities of the municipality, which is of central importance in relation to environmental and long-term sustainability.

To measure the trajectory of development with reduced soil consumption the indicator; amount of square meters of unspoiled soil consumed, will be analyzed. More specifically, unspoiled soil is land that has not been compromised such as green or agricultural areas. Thus, by measuring the soil consumption of uncompromised soil with the aim of reducing it, the municipality can monitor how the city's development and expansion is impacting the environment and, therethrough, enhance sustainability and ecosystem quality.

3.3.5.5. S: Civil society and a vibrant city

S1: Create safe, inclusive, and diverse public areas

The fifth and last CDF correlates to the strategic objective by addressing targets to make Linköping vibrant and attractive from a societal and economic perspective. The first target, creating safe and inclusive public areas, aims to enhance community engagement and cohesiveness to foster a vibrant society. This target will be assessed through two criteria; (Ac.S1.1.) quality and safety of spaces, and (Ac.S1.2.) availability of diverse functions and uses. Thence, by creating diverse and inclusive public spaces, public gathering and community spaces will be incentivized. This is important in order to foster a vibrant and safe society. Namely, by enhancing community cohesion and engagement social capital can increase and enhance perceived and actually safety parameters (Slocum et al., 2024).

The assessment criteria will thereby be indicated through analyzing the; (i) qualitative data of safety protection, and (ii) percentage of diverse uses and function. Thus, to enhance communities and create safe public areas, the first indicator aims to conduct qualitative surveys of perceived safety of citizens and users of public spaces. This is significant in order for the municipality to identify and mitigate safety issues in relation to public areas and the overall city. The second indicator aims to measure the percentage of diversity functions and uses. This entails

ensuring the opportunity and space for different community activities and groups to be accessible and supported by the municipality and its citizens. By investing in and improving the safety and inclusiveness of the city, the city can develop vibrant communities and present a more attractive city.

S2: Sustainable public transport and mobility

Furthermore, the second target, sustainable public transport and mobility, bridges the social and ecological aspects of sustainability by ensuring the accessibility and sustainability of public transit and mobility infrastructure. This is encompassed by the assessment criteria (Ac.S2.) availability of active mobility infrastructure and possibility to move using public services, such as public transit and bike lanes. Effective, accessible and sustainable transport is significant due to its impact on citizens' everyday life, in terms of travel time, as well as the city's environmental qualities, such as reducing carbon footprint in relation to increased public transit and bike usage.

This assessment criteria will incorporate three indicators, namely; (i) data about quality of existing bike paths, (ii) distance (square meters) between residential buildings and transport stations, and (iii) travel time between residential areas and work places. The first indicator will be utilized in order to maintain the existing bike paths' standard, which are heavily used as indicated by the interview with civil society, highlighting the central importance of bike transportation in the city. The data, analyzing bike path quality, network and correlating infrastructure, such as pumps, street lights and bike stands, will enhance municipal information for further maintenance and expansion of bike lanes. The second indicator aims to address the accessibility of public transport, in terms of distance between residential buildings and public transport hubs. Thus, by identifying current and potential areas for increasing proximity to public transport the public transit ridership is likely to increase, further contributing to reducing GHG emissions within the municipality. Lastly, the third indicator aims to identify the effectiveness of travel time as well as accessibility of services, economic and employment opportunities. This will be done by measuring travel time from residential areas to work places to identify areas of improvement of both efficiency and accessibility of different transport infrastructure

S3: Improve the business centre, and extend ways to attract and enhance entrepreneurship and business activities

Another aspect of creating a vibrant and attractive city entails economic sustainability. This is addressed through the target; improve the business center and enhance entrepreneurship and business activities. This will be ensured through the assessment criteria (Ac.S3.) increase policy to enhance economic activity. This assessment criteria aims to address policy and institutional investment for enhancement of economic activity and opportunities.

This assessment criteria, will therethrough be indicated by analyzing and monitoring the percentage of working inhabitants and employees within the city. Hence, by measuring the proportion of the city's population working within local businesses and industries the economic sector can be assessed. By assessing aspects of the economic sector, matters such as local economic activity and employment rates can be incorporated into further development prospects. The specific focus on local businesses and industries therefore provides an indicator of long-term economic stability as well as creates further enhancement and opportunities for entrepreneurship. This indicator can

also contribute to identifying certain sectors or proportions of the population seeking income elsewhere or further improvement of the market and businesses within Linköping.

S4: Offer for more affordable and accessible housing

The last target to offer more affordable and accessible housing. This is of central importance to the citizens, temporary and permanent, within the city. Based on the civil society interview one central concern within Linköping are high accommodation prices. This target will be assessed through (Ac.S4.) home availability and affordability. In order to measure this, two indicators will be used: (i) Amount of square meters of new buildings, and (ii) policy price caps on rent for students and low-income population.

The first indicator aims to address the housing shortage within the city. Therefore, by measuring the amount of square meters of new accommodation units, new housing development can increase supply and lower demand for housing. This development can entail developing new buildings as well as repurpose old buildings and land to create new apartments and houses. The second indicator aims to address the affordability of accommodation in Linköping by introducing rent price caps for student and low-income populations. This is a significant tool in order to maintain students and low-income people within the city and contribute to long-term sustainable socio-economic development.

3.3.6 Strategic Options

With the identified strategic objective, CDFs, targets, assessment criteria and indicators, strategic pathways can be drawn in order to illustrate how to reach the desired future vision (Partidario, 2012). These pathways, so called strategic options, have different approaches and trajectories towards the strategic objective. More specifically, strategic options aim to address all the CDFs, to different extent, utilizing either policy or technical based strategies (Balfors, 2026). For Linköping municipality, 9 strategic options have been drafted, based on the target as well as associated assessment criterias, illustrated in figure 9, and will be further disclosed.

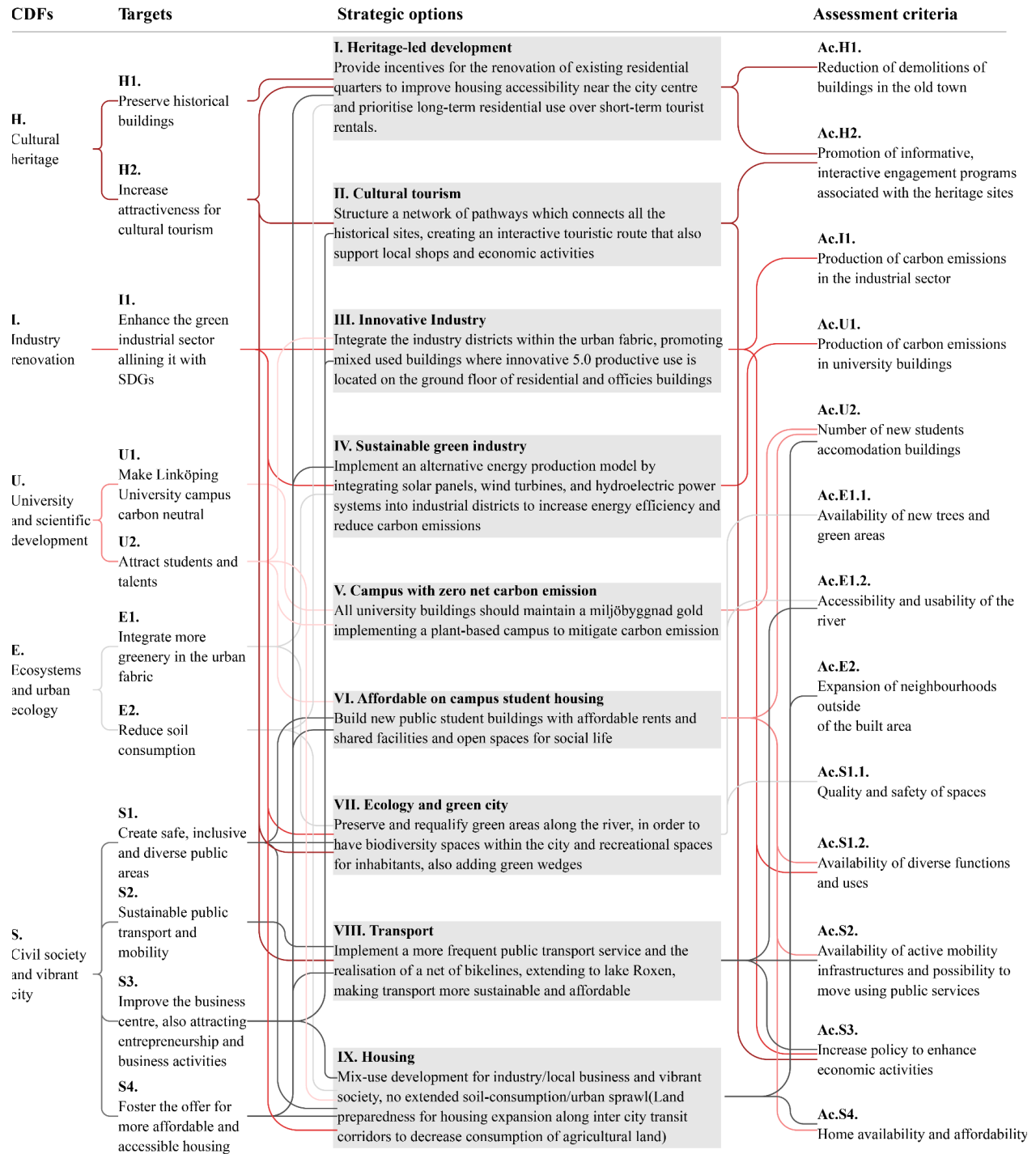


Figure 9: Strategic options

3.3.6.1. SO I. Heritage-led development

The first strategic option aims to encompass the first CDF (cultural heritage). Thus, strategic option I aims to *provide incentives for the renovation of existing residential quarters to improve housing accessibility near the city centre and prioritise long-term residential use over short-term tourist rentals*. This approach will therethrough

address targets H1, H2, E2 as well as S4, by preserving historical buildings, increasing cultural tourism, and offer more accessible housing, respectively.

Infrastructure and architecture is central to a city's identity and carries significant historical meaning, in terms of urban form, traditions and cultural identity (Di Figlia, 2016). Nonetheless, cultural and historical buildings are significant for a city and its inhabitants, however, they do demand maintenance and restructuring. With this maintenance demand possibilities of creating development for cultural tourism and heritage-led development are presented (Di Figlia, 2026; Femenias et al., 2023). Repurposing cultural buildings can entail both restructuring of electricity and water networks or repurposing spaces from historical use towards for example accommodation. The latter presents an opportunity for Linköping through preserving the cultural identity of the city whilst combining it with the need for accessible and affordable housing, as expressed in the Linköping municipality masterplan (2010) and both interviews. This strategic option can thereby be inspired by heritage-led sustainable development initiatives, which focuses on integrating community values and cultural heritage with cultural tourism and circular economy practices (Candeloro & Tartari, 2025). Adapting a heritage-led development plan in Linköping through repurposing and maintenance of cultural buildings can provide not only long-term quality of cultural buildings, and increased cultural tourism and economic activities, but it can also contribute to increasing accommodation provision in the city, also avoiding the construction of new buildings in unspoiled fields.

Nonetheless, when repurposing old buildings it is important to consider the buildings in a socio-historical context and how this change will impact local citizens as well as the buildings (Femenias et al., 2023; Campo-Ruiz, 2012; SPAB, n.d.). The significance of heritage change is inevitable, since modernization and development is a continuous process, whereas the legacy of cultural buildings should be preserved, the repurposing of buildings need to be done in a careful manner and with a holistic approach (Hupkes, 2024; SPAB, n.d.).

Another focus point in this strategic pathway is to reduce short-term rentals within the housing market to further enhance accommodation accessibility and affordability. This is due to, as for example in Stockholm, where private real-estate actors and Airbnb online rentals have created a significant change of the housing market in the city (Adamiak & Marjavaara, 2024). Namely, short-term rentals, such as Airbnb, can contribute to tourism accessibility but, however, increasing housing market prices, reduces availability of accommodation, and increases gentrification, through displacement of low-income populations, as it is primarily driven by private and market actors (Adamiak & Marjavaara, 2024; Campo-Ruiz, 2023). This gentrifying effect can also be evident within repurposing of old buildings.

Nonetheless, by proposing this strategic option of heritage-led development driven by the municipality, and associated stakeholders, a more centralized control of the housing market, prioritizing long-term rentals, can counteract housing market forces. The strategic option also fosters cultural tourism and can present opportunities in terms of cultural heritage preservation and revitalization (Di Figlia, 2016). This can further enhance the attractiveness of the city through maintenance of the city center and its cultural and historical identity. Moreover,

this strategic option also integrates central legislation of cultural protection of buildings (K-märkning) and development should be performed with regards to this legislative framework.

3.3.6.2. SO II. Cultural Tourism

In connection to the first strategic option, enhancing the cultural heritage of Linköping, the second strategic option focuses on promoting cultural tourism in Linköping. Thus, by *structuring a network of pathways which connects all the historic sites, creating an interactive touristic route that also supports local shops and economic activities*, this strategic option will aim to address targets; H2, S1 and S3.

To begin, tourism plays a significant role for a city or a region's economy, particularly for smaller cities (IBEF, 2026; Tillväxtverket, 2014). A large part of tourism comes from cultural tourism, namely where cultural sites or experiences are the central objective for tourists. Cultural tourism is one of the primary sorts of tourism in Sweden and is incentivized by many collaborative programs between national and regional actors (Visit Sweden, n.d.). This is of relevance for Linköping with its rich supply of art and historical activities as well as cultural monuments (Visit Linköping, n.d.).

Moreover, cultural tourism is significant for Linköping's development, thereby, the second strategic option aims to promote and optimize sustainable cultural tourism. This is proposed with a suggestion, inspired by the Cultural Routes of the European Program (Council of Europe Portal, 2026). The cultural routes of Europe is an initiative which provides networks and services connecting central cultural and historical sites throughout Europe. This increases the accessibility and cultural exchange of tourism. This extensive and widerange program is thereby proposed on a much smaller scale to enhance Linköping's cultural tourism. Thus, by identifying central sites for cultural heritage and local economic activity a road or walking paths, with interactive information, can enhance the accessibility and attractiveness of the tourist sector in a feasible and sustainable manner.

To further enhance the sustainability of this strategic option, and incorporate aspects of the civil society CDF, another Italian strategy is suggested (Monaco et al., 2026). Namely, the incorporation and investment in social enterprises to not only enhance community-led tourism and local economy but also to foster social inclusion and sustainability (Monaco et al., 2026). Since, community-led social enterprises can increase the interactive and locally embedded tourism activities, whilst also increasing the economic opportunities presented to the residents of Linköping. Nonetheless, increasing community collaboration and social enterprises is dependent on institutional support and adequate infrastructure (Monaco et al., 2026). To mitigate these limitations, the strategic option presents an opportunity for the municipality and associated institutions to provide support and contribute to building the connective cultural tourism path, whilst enhancing sustainability, local economy and social cohesion.

3.3.6.3. SO III. Innovative Industry

To further enhance the sustainability of Linköping, a green industrial sector will be the central objective for the third strategic option. So, strategic option 3 aims to *integrate the industry districts within the urban fabric, promoting mixed-use buildings where innovative 5.0 productive use is located on the ground floor of residential and office buildings*. Through integration of industry within the urban fabric, this strategic option, directly and indirectly, covers targets I1, U2, and S3.

The initial process of reaching this strategic option stems from looking at land-use maps and identifying industrial areas. However, with modernization the municipality aims to repurpose some of the old industrial areas, such as Hackefors (Linköpings Kommun, 2025). Within this area the municipality aims to modernize old industrial areas and buildings to align with contemporary standards of efficiency and sustainability, through increasing density and incorporating new activities (Linköpings Kommun, 2025). Nonetheless, this strategic option offers a complimentary pathway to the municipal objectives by incorporating and integrating industry into the urban fabric. Additionally, this integration will be facilitated by new development and redevelopments focusing on mixed-used buildings.

Mixed-use buildings is a planning and building strategy which aims to have residential areas, work spaces and central services within the same building or within close proximity (Bellander, n.d.). Thus, by creating productive spaces at the ground floor in residential or office buildings a more innovative business environment can be enhanced with close connections to offices and market actors. Additionally, by combining service and residential areas a more accessible and vibrant society can be fostered through employment opportunities and services within walking distance. By bridging the industrial, business and social, industrial activities located in the urban fabric can be facilitated, which will leave new spaces in the industrial areas for repurposing to create a further integrated and accessible city. This opportunity has been strategized in Singapore, where system integration enables the transformation of industrial landscapes utilizing circular and greening strategies (Samant, 2026).

To enable the transition towards an integrated industry, at an applicable level, the incorporation of an Industry 5.0 approach is suitable. This is an approach promoted by the a.European Union (n.d.). This approach encompasses placing industry at the forefront of technical, digital and green transitions. Namely, that industrial precision should maximize sustainable and technical solutions in order to foster worker health and create industries for society (ibid). Additionally, this creates incentives for innovative solutions within green transitions and AI, such as the city's digital twin.

Notwithstanding, this strategic option can foster an integrated urban form as well as incentives and investment into innovative solutions and modern industries, through adapting Industry 5.0 approaches. By integrating industry, focusing on renewal and modern industries, this strategic option offers further future development pathways through for example enhancing the collaboration between the university and industrial sector (a.Liu, n.d.). By enhancing the collaboration, positive effects can be translated through students gaining practical experience, enhanced R&D, innovative solutions, as well as a more integrated society, connecting the university to the rest of the city, as mentioned in the interview with civil society.

This strategic option presents valuable opportunities for the municipality by integrating industry, fostering innovative environments, increasing accessibility and interconnectedness of different sectors within the society. Nonetheless, considering the impact of industry on the environment this strategic option should be aligned with regulations in regards to residential safety and the environmental code of Sweden (Boverket, 2015; a.Sveriges Riksdag, 1998).

3.3.6.4. SO IV Sustainable Green Industry

Continuing, the fourth strategic option intends to implement green energy production in Linköping and its industrial sector. Specifically, strategic option IV, sustainable green industry, focuses on *implementing an alternative energy production model by integrating solar panels, wind turbines, and hydroelectric power systems into industrial districts to increase energy efficiency and reduce carbon emission*. By utilizing renewable energy sources the fourth strategic option will encompass target I1, and indirectly E1 and S4.

Strategic option IV emphasizes the usage of renewable energy, and therethrough phasing out fossil fuels energy consumption. This is accentuated due to the central contribution of GHG emissions from the industrial sector, since this sector contributes to around 30% of all emissions in Sweden, primarily from iron and steel industries (Naturvårdsverket, 2025). The incentive and investment in renewable energy proposed by strategic option IV can play a crucial part in Sweden's ambition for a sustainable and a net-zero emission target and the SDGs (UN, n.d.). This strategic option will be enacted in symbiosis with the increased investments and ambitions of both the Swedish government and the EU towards an effective green energy transition (Tillväxtverket, 2026; Regeringskansliet, n.d.). The green transition in Linköping's industrial sector will also contribute to the production and services, from Linköping, to be more competitive on a global market, which can enhance further growth, development and innovation in the city (b.EU, n.d.).

Linköping currently has one of the country's largest solar cell parks, and is planning another one in Bjursby, close to Linköping (Tekniska Verken, n.d.; Greengo Energy, n.d.). Nonetheless, with the strategic option a further enhancement of the renewable energy sectors in Linköping is complemented by a strategy of hybrid renewable systems. This entails integrating different renewable energy sources, such as solar, wind and hydropower, to create a more stable energy grid and increase cost-effectiveness of energy infrastructure (Hassan et al., 2023). This strategic option can increase the city's sustainability production and consumption as well as provide avenues for cheaper energy prices and reduced fossil fuel dependency (Ortega et al., 2026). This can have positive repercussions on energy costs of buildings, taking advantage of the local electric energy and of the possibility of creating electric communities.

Lastly, to ensure the technical green energy transition within the industrial sector, policy tools can be further implemented to enhance the effectiveness and sustainability of the sector. Thus, by complimentary policy action by the municipality, through for example stricter monitoring of emissions and percentage of renewable energy use in combination with fines for exceeding said percentages, can enhance and incentivise the green transition.

3.3.6.5. SO V. Campus with zero net-carbon emission

Furthermore, to align with Sweden's carbon neutral goal this strategic option aims to transition one of the largest institutions within the municipality, the university, to have zero net-carbon emissions. This will be reached by *all university buildings maintaining a miljöbyggnad gold standard, and implementing a plant-based campus to mitigate carbon emissions*. This strategic option will thereby address targets; U1 and U2.

By incorporating national sustainable building standards, Linköping University will adhere to guidelines and practices at a national level to ensure the sustainability and carbon-neutrality of the campus. Miljöbyggnad Certification, by Sweden Green Building Council (n.d.), is the most used sustainability certification for Swedish buildings. The certification can be applied to most buildings, including schools such as Linköping University Campus. The certification is based on 15 indicators which address different sustainability aspects such as environmental impact, energy consumption, and circularity of the building (ibid). This thereby presents as an accessible and official tool to ensure the sustainability of the campus.

Linköping University buildings currently have some buildings with miljöbyggnad gold certification, whereas others have silver. Nonetheless, the university actively recognizes the significance of a sustainable campus (b.LiU, n.d.). Thereby, by providing further municipal incentives and support to reach a sustainable campus this strategic option provides further guidelines to reach this target. The increase of sustainability of the campus, in combination with other strategic options, will create a university more prone to attract new students.

In combination with miljöbyggnad gold certification of campus buildings, is the ambition for a plant-based campus to further reduce emission. Research by Xu and Lan (2016) indicates that a plant-based campus reduces the carbon footprint. This is further supported by Zhang et al.'s (2024) study which investigated the environmental impact of plant-based default options at different university institutions. The results indicate a significant reduction in CO2 emissions, land-use consumption, nitrogen use, and phosphorus use. Thus, by creating an integrated strategic option with miljöbyggnad standards in combination with approaches such as plant-based campuses Linköping University campus can contribute to reaching climate goals and promote a more sustainable university environment.

3.3.6.6. SO VI. Affordable on campus student housing

The sixth strategic option is derived from the interview with civil society as well as thematic trends within the Linköping masterplan, namely, the price and availability of housing, more specifically student housing. As Linköping is one of Sweden's largest university cities the availability of housing is of central importance for the city's long-term development. The strategic option will address the matter of *building new public student buildings with affordable rents and shared facilities and spaces for social life*. This strategic option thereby covers targets U2, S1, and S4.

Student accommodation, as mentioned in the interview with civil society, is relatively expensive in relation to Swedish student loans. Linköping's primary student accommodation providers are Studentbostäder and Heimstaden where prices for corridor rooms and one room apartments range from 4 000 kr to 6 500 kr, with a current low supply options (Studentbostäder, n.d.; Heimstaden, n.d., date of consultation: 18.02.2026). Thereby, the capacity of the city to provide accessible and affordable student accommodation would increase with the provision of new, or repurposed, student accommodation.

The new student accommodation would be equipped with shared facilities and community spaces. This would, as in other student cities, contribute to fostering a cohesive and interactive academic and social environment. The spaces could also further catalyze student association and unions. Since, student associations do not only contribute to community coherence but also play a significant part in shaping the university and promoting democratic and representative processes within academia (UNESCO, 2024). The impact of student unions is not only to impact academia, but can also increase social capital, contributing to a vibrant civil society, as well as impact housing prices (Stockholm University, n.d.). Thus, this strategic option can foster a positive feedback loop between housing, academia and student associations as well as civil society.

Furthermore, this is a policy-based strategic option which aims to foster a sustainable and equitable housing market for student accommodation. This can thereby imply policy tools such as price-cap regulation on student accommodation rents, as well as municipal support and investment in creating new and sustainable student accommodation units, through both producing and repurposing buildings.

3.3.6.7. SO VII. Ecology and green city

Strategic option 7, ecology and green city, objectifies to *preserve and requalify green areas along the river, in order to have biodiversity spaces within the city and recreational spaces for inhabitants, as well as incorporating green wedges*. This strategic option therethrough encompasses targets: H2, I1, E1, and S1.

Firstly, the green areas along the river should be preserved, at good quality, or requalified. Requalification of green areas entails increasing the quality of green areas, in combination with introducing more urban green spaces, to ensure the health of the ecosystem and biodiversity (Jones, 2025). The river in Linköping flows throughout the central part of the city and is thereby affected by the activities in and around the city. Thus, by ensuring the health of green spaces, as well as the river, through for example evaluating ecological networks and investing in maintenance of biodiversity and green areas (Szulczewska et al., 2014). This further serves to address matters of the municipality's current frameworks and climate adaptation objectives, as highlighted by the interview with the municipal planner.

Nonetheless, another approach to ensure an ecologically valuable and green city is through the introduction and implementation of green wedges. Green wedges have been implemented in Stockholm and are made up of continuous wedges of undeveloped land or green urban areas (Esmail et al., 2022; Region Stockholm, 2025). The

green wedges function as a strategy to incorporate and integrate green areas within densely populated urban areas, with the wedges in the shape of a star towards the center of the city. The central strengths of the green wedges is its ability to create continuous corridors of green areas which allows connectivity and mobility of ecosystems, flora and fauna (ibid). Apart from the significant ecological value of the green wedge system, it also protects and generates green spaces for cultural and recreational purposes (Esmail et al., 2022; Region Stockholm, 2025).

This strategic objective creates multiple reinforcing positive impacts through protecting green areas along the river and creating a green city, through for example the introduction of green wedges. Primarily, the focus on protecting and requalifying green areas biodiversity and ecological values will be enhanced (Region Stockholm, 2025; Paudel & States, 2023). A greener and healthier urban area also creates opportunities for green, and blue, infrastructure and nature-based solutions, which contributes to climate adaptation and resilience within the city (ibid). Thus, by investing in green requalification, Linköping can further enhance its urban environmental qualities through ecosystem services. Furthermore, green areas can also contribute to human health by reducing air pollution, urban heat islands as well as providing spaces for social gathering and recreation (Vanhöfen et al., 2025; Cyprianska & Nezlek, 2018; EEA, 2024). Nonetheless, to implement this strategic option, development and actions should be done in line with Sweden's environmental code. This strategic option thereby has the opportunity to address both the green, attractive and vibrant aspects of the strategic objective.

3.3.6.8. SO VIII. Transport

Continuing, strategic option 8 aims to address aspects of sustainable and accessible transport and mobility within Linköping. More specifically, strategic option 8 will *implement a more frequent public transport service and the realisation of a net of bikelines, extending to lake Roxen, making transport more sustainable and affordable*. This strategic objective will, thereby, address targets H2, S2 and S4.

This strategic option aims to increase public transport frequency and accessibility, which can increase public transportation ridership, as a complementary local action in combination with the regional Ostlänken development. As Ingvardson and Nielsen (2018) indicated, in their research on public transportation ridership in 48 European metropolitan areas, is that the extension of public transport networks have a positive effect on public transport ridership. Whereas, cities with higher socio-economic inequality have a lower ridership, thus pointing to a correlation between economic impacts such as price and spatial and social exclusion, which has also been evident in transport research in Brazil (Turbay et al., 2016).

Thus, by increasing public transport ridership, in combination with expanding bike paths, the city's transport and mobility sector can vastly enhance Linköping's sustainability performance, enhance environmental quality, and contribute to Sweden's zero-net GHG emission goal (Naturvårdsverket, 2023). The extension and sustained maintenance of bike paths is significant for the city as it is one of the central modes of transport, particularly for students. The sub-objective of this strategic option is to extend bike lanes to lake Roxen, as highlighted in the civil society interview. The interviewee emphasized the lack of infrastructure to the lake, which particularly limits social

and recreational activities during the summers. Hence, by investing in extending transport and mobility networks towards central recreational areas in and around Linköping, this strategic option can further enhance and fuel a vibrant society.

Nevertheless, expanding transport infrastructure, such as public transport and bike lanes, demands investment which makes it particularly relevant for the municipality. However, as the strategic option regards transport it also requires collaboration with stakeholders, and their legal frameworks, such as Trafikverket and Östgötatrafiken. Moreover, when investing and creating infrastructure it is necessary to consider the long-term impact of infrastructure. Namely, by building a road or bikelane, it will most likely persist for many years (Birch, 2016). Thus, when adapting for this strategic option it is important to incorporate future planning objectives, and climate change impacts, within the planning and implementation process. To conclude, this strategic objective demands substantial investment and planning, nonetheless, it also presents opportunities to enhance social and environmental sustainability, as well as contribute to the SDGs and strategic objective through foresting a vibrant and attractive city.

3.3.6.9. SO IX. Housing

The last strategic option is focused on addressing the matter of expensive and inaccessible housing in Linköping. More specifically, *creating mix-use development for industry/local businesses and a vibrant society, without further soil consumption/urban sprawl, and instead creating land preparedness for housing expansion along inner-city transit corridors to decrease consumption of agricultural land*. This strategic option will thereby target: S1, S3, S4, I1, U2, and E2.

To begin, the ninth strategic option focuses on mixed-use developments. Mixed-use development is a planning approach which aims to integrate activities and services in the ground floor of residential buildings, in combination with creating housing in close proximity to services, jobs and public transport (Boverket, 2024). Utilizing this housing approach carbon emissions from transport will be reduced as well as increased local social and economic activity. Nonetheless, it is important to consider contextual and long-term aspects, consequences and visions when constructing mixed-use development in order for it to foster socio-economic activity and sustainability (Boverket, 2024). Thus, to complement complex planning issues, in terms of urbanization and urban growth trends, this strategic option suggests the incorporation of land preparedness within the inner-city of Linköping. Land preparedness entails creating sufficient infrastructure and creating valuable preconditions for creating housing development.

To further integrate the identified targets, this strategic objective emphasizes the value of creating housing development through repurposing of already constructed buildings or consumed land. This is proposed in order to decrease soil consumption and urban sprawl. Since, urban sprawl, as a result of urban growth, can create new obstacles such as increased car-dependency and negative ecological impacts (Glaeser & Kahn, 2004; Peiser, 2001). However, with the strategic option, soil consumption of agricultural land can be reduced as much as possible, while

the city center is repurposed and regenerated into a more vibrant and accessible city. This thereby creates an alternative solution to the challenges highlighted by the interview with the municipality.

Withal, this strategic option, and proposed housing developments, still presents risks. These risks are associated with the increased development in the city center as well as repurposed or newly developed housing which will be attractive on the housing market. With large investments particularly in central areas the risk of gentrification is increased, increasing the prices on the housing market and thereby pushing low-income populations towards the periphery (Thörn et al., 2023; Tunström & Wang, 2019). The impact of gentrification can contribute to further segregation, as was mentioned as a challenge by the interview with the municipality. However, policy and municipal mitigation measures are available such as rent caps to reduce market forces in combination with a mixture of tenure types. Thus, by incorporating different tenure types, gentrification can be counteracted by not allowing only owned apartments in one area, which often implies a certain income-range within the population, and instead creating and fostering a mix of both tenure types and population groups can contribute to a vibrant society (Wimark et al. 2021).

Furthermore, some central regulatory and legislative frameworks this strategic option should adhere to is the Planning and Building Act (PBL) in Sweden. Additionally, with adapted mitigation measures as well as ensuring sustainability objectives and targets are regarded this strategic option can contribute to creating more housing in combination with a more vibrant and accessible Linköping.

3.3.7 Opportunities and risks

The presented strategic options provide alternative strategic pathways for Linköping to reach the strategic objective of this strategic thinking SEA. Despite the effort to adapt contextualized and holistic solutions, the strategic options present both risks and opportunities. The identified risks and opportunities, for each strategic option, have been evaluated in correlation to each assessment criteria. The results from these analyses are presented by the final matrix, table 5 (*see chapter 3.3.8*), as well as a more elaborate written explanation in Table 6 (*see chapter 3.3.9*). Nonetheless, a further motivation of the opportunities and risks will be disclosed in relation to the recommended strategic options (*see section 4*).

3.3.8 Final matrix

Assessment criteria*	Strategic options								
	I	II	III	IV	V	VI	VII	VIII	IX
Ac.H1. Reduction of demolitions in the old town	↑	↗	→	NA	NA	NA	↗	↘	↗
Ac.H2. Promotion of heritage sites	↑	↑	NA	NA	NA	↗	NA	↗	↘
Ac.I1. Production of carbon emissions (Industry)	↗	NA	↑	↑	NA	NA	↓	↓	↘
Ac.U1. Production of carbon emissions (University)	NA	NA	NA	↗	↓	↓	↓	↓	↘
Ac.U2. Number of new student accommodations	↗	NA	↘	NA	↑	↑	NA	→	↑
Ac.E1.1. Availability of new trees and green areas	↘	NA	NA	↓	↗	↗	↑	↘	↘
Ac.E1.2. Accessibility and usability of the river	NA	NA	NA	↓	→	↗	↑	↑	NA
Ac.E2. Expansion of neighbourhoods (soil consumption)	↗	→	→	↓	→	↗	↓	↗	↑
Ac.S1.1. Quality and safety of spaces	→	↗	↗	↘	NA	↗	↑	↘	→
Ac.S1.2. Availability of diverse functions and uses	↗	↗	↑	NA	↗	↑	↗	↗	↗
Ac.S2. Availability of active and public mobility	NA	↗	NA	↗	NA	↑	↘	↑	↗
Ac.S3. Increase economic activity	NA	↑	↑	NA	NA	↗	↗	↑	↗
Ac.S4. House availability and affordability	↗	↘	↓	↗	↗	↑	NA	→	↑
Total points gained	+4	+5	+6	-2	+3	+12	+2	+2	+6

* Some assessment criteria have been shortened. For a more specific description refer to Table 4.

References of the icons and associated value

Strong risk	Risk	Neutral	Opportunity	Strong opport.	Not applicable
↓	↘	→	↗	↑	NA
-2	-1	0	+1	+2	0

References to the number of rows

I. Heritage-led development Provide incentives for the renovation of existing residential quarters to improve housing accessibility near the city centre and prioritise long-term residential use over short-term tourist rentals.	II. Cultural tourism Structure a network of pathways which connects all the historical sites, creating an interactive touristic route that also supports local shops and economic activities.	III. Innovative Industry Integrate the industry districts within the urban fabric, promoting mixed used buildings where innovative 5.0 productive use is located on the ground floor of residential and office buildings.	IV. Sustainable green industry Implement an alternative energy production model by integrating solar panels, wind turbines, and hydroelectric power systems into industrial districts to increase energy efficiency and reduce carbon emissions.	V. Campus with zero net carbon emission All university buildings should maintain a miljöbyggnad gold, implementing a plant-based campus to mitigate carbon emission.
VI. Affordable on campus student housing Build new public student buildings with affordable rents and shared facilities and open spaces for social life	VII. Ecology and green city Preserve and requalify green areas along the river, in order to have biodiversity spaces within the city and recreational spaces for inhabitants, also adding green wedges	VIII. Transport Implement a more frequent public transport service and the realisation of a net of bikelines, extending to lake Roxen, making transport more sustainable and affordable	IX. Housing Mix-use development for industry/local business and vibrant society, no extended soil-consumption/urban sprawl(Land preparedness for housing expansion along inter city transit corridors to decrease consumption of agricultural land)	

Table 5: Final matrix and its legends

3.3.9 Monitoring and Follow-up

The land use sector of Linköpings municipality requires a SEA to support sustainable development. It is further needed to guide its long-term investments. To ensure a sustainable future, monitoring and follow-up represents essential final steps in the SEA process. Effective monitoring is necessary to determine whether the assessment has achieved its intended outcomes, and to further identify any unforeseen environmental effects.

To monitor the land-use changes, several approaches can be implemented. A usual method is the comparison of spatial areas over time using mapping techniques such as Geographic Information Systems (GIS). GIS enables the identification and analysis of land use activities and developments. GIS further includes residential and commercial expansion. Mapping can also be employed to track changes in transportation infrastructure and agricultural land. This spatial analysis should be the main monitoring strategy for Linköoings municipality. It allows for a systematic evaluation of development patterns, and provides an insight of the environmental impacts (Maguire et al., 2005).

Furthermore, as Linköping's municipality is expected to grow in line with its comprehensive plan, it is important to ensure that the development aligns with the needs of current inhabitants and other relevant stakeholders within the municipality. This could be achieved by establishing annual public feedback opportunities. These opportunities include hearings, workshops, surveys, and digital mailboxes, which would provide valuable insight from the public perspective. In addition, considering the risks associated with the proposed strategic options, a structured table of monitoring and follow-up measurements has been presented in the table below.

SO	Risks	Opportunities	Monitoring	Indicators
CULTURAL HERITAGE				
SO I: Heritage led development	(i) Repurposed accommodation → increased housing prices → displacement of low-income residents.	(i) Increase cultural values within the city, which can contribute to increased tourism (national and international) → contributing to the local economy.	(i) The County Administrative Board (Länsstyrelserna)	(i) Monitor the housing prices/share of permanent/tourist residents
	(ii) Disturbances to public transport and noise pollution.	(ii) Repurposed housing, and cultural preservation, can create attractive accommodation → attract new residents, increasing long-term economic and social development.	(ii) Samhällsbyggnadsnämnden of Linköpings Municipality. (Spatial planning authority)	(ii) Monitor through GIS, comparing spatial development
	(iii) Can interfere or clash with the aspirations to increase housing and new buildings in the city		(iii) Environmental code (1990:808, Chapter 6). Cultural Environmental act (1988:950), and Planning and building act (2010:900).	
SO II: Cultural tourism	(i) Overtourism: A lot of tourism, might disturb the locals	(i) Short-term residents becoming residents due to an attractive and vibrant society (e.g. not only study there through a vibrant civil society and thriving local economy)	(i) The County Administrative Board (Länsstyrelsen). Culture and Leisure administration	(i) Visitor density, and public transportation load.
	(ii) Economic over-dependency and vulnerability in the tourism sector.	(ii) Improve the marketing and create a competitive city branding → improve economy and a vibrant civil society	(ii) Environmental Code and Directive 2001/42/EC.	(ii) Measure loss of identity through mapping.
	(iii) Loss of identity: cultural expressions are often turned into commercial, tourist-friendly performances	(iii) Cultural and historical knowledge increase through preservation of old buildings and cultural heritage		
	(iv) Creating pressure and demand for further expansion of tourist activities/attractions which can compete with interests of no-land-take planning.			
INDUSTRY (GREEN) RENOVATION				
SO III: innovation industry	(i) Demands space, and in combination with no soil consumption targets, this risks displacing current residents.	(i) Improve new ways of production and incentives for innovation. Enhancing technological and digital development, within an environment close to other actors, consumers and the market.	(i) Environment and Spatial Planning Authority (Miljö- och Samhällsbyggnadsförvaltningen) of Linköpings Municipality.	(i) Buffer-zones and distances. Set pollution levels/thresholds.
	(ii) Risk of pollution (chemical and noise) if no monitoring or standards are implemented	(ii) Contribute to innovating/redesigning functions and parts of the campus and other central infrastructures of the city	(ii) Swedish environmental protection agency (Naturvårdsverket).	(ii) Measure number of annual households moving out in regards to industry areas.
	(iii) Risk of polluting waters and contaminating soil by implementing hydropower, affecting the ecosystem	(iii) Innovative and sustainable green solutions for new green infrastructure and parks, which can indirectly contribute to societal and recreational values	(iii) County administrative board (Länsstyrelsen), SGU (Swedish geotechnical investigations)	

SO IV: Sustainable Green industry	(i) Green washing	(i) By reducing emission and integrating green roofs and urban forestry within industrial parks, industrial areas could become a part of climate adaptation strategies, rather than just a source of pollution. Could turn grey industrial areas to ecological networks through habitat restoration, and buffer zones.	(i) Environment and Spatial Planning Authority (Miljö- och Samhällsbyggnadsförvaltningen) of Linköpings Municipality.	(i) Sustainability reporting. Track pollution, and carbon footprint.
	(ii) Some industries might not have the resources or capacity to implement a quick green transition	(ii) By utilizing renewable energy the production and products will be more adapted to future possible environmental regulations as well as the products presented on the international market will become more attractive due to its sustainability.	(ii) Swedish environmental protection agency (naturvårdsverket)	(ii) Circular Economy. Measure greenhouse gas emission by percentage
	(iii) Risks increasing the price of local products creating (due to increased investment in renewable energy transitions) a market more beneficial for external products .	(iii) With an increase in combined renewable energy systems, the energy grid will become more stable and resilient		(iii) Adapt Lifecycle assessment (LCA).
UNIVERSITY AND SCIENTIFIC DEVELOPMENT				
SO V: Campus with zero net carbon emission	(i) Increase of institutional pressure as well as investment demand, creating financial and operational obstacles.	(i) Increase the attractiveness of the campus by ensuring a long-term sustainable impact on the environment.	(i) Samhällsbyggnadsnämnden (Spatial planning authority). Stadshuset (town hall). Akademiska hus. Sweden Green Building Council	(i) Number of certified environmental buildings; bronze, silver, gold.
	(ii) Disturbance from rebuilding and maintenance work	(ii) Entails new sustainable solutions and infrastructures to improve climate adaptation and resilience, both through nature-based solutions and sustainable transport.	(ii) Planning and Building Act (2010:900). Swedish National Board of Housing. BBR - Boverkets Building Regulations (2011:6). Energy performance cert. (Boverket, 2023).	(ii) Monitor disturbance from rebuilding and maintenance by implementing noise monitoring systems, keep below 65 dB
SO VI: Affordable on campus student housing	(i) Demands resources and can create conflicts with other objectives, such as no soil consumption	(i) Increased incentives for university development present opportunities for increased student engagement, cultural and knowledge exchanges, innovative solutions and a vibrant city	(i) Linköpings University	(i) GIS-Mapping as an indicator to track expansion of land use.
	(ii) Can further fuel the division between students and the civil society, since accommodation is built on campus	(ii) Incentives to invest in student association can contribute to further incentive of more recreational and park areas which would further contribute to a vibrant and sustainable society	(ii) Planning and Building Act. Akademiska Hus.	(ii) Measured green space per student, available parks, open space
ECOSYSTEMS AND URBAN ECOLOGY				
SO VII: Ecology and green city	(i) Risk interfering with with the no-land-take policies	(i) An increase in green areas and ecosystems presents further opportunities to improve air quality and reduce noise pollution	(i) Environmental Code and Directive 2000/60/EC.	(i) Water and air quality index.
	(ii) Demands further maintenance and investments to maintain the quality of greenery and ecosystems, despite increased use, climate change, and population growth.	(ii) A greener city can improve the livability of the city and increase the satisfaction of the inhabitants		(ii) Biodiversity index
		(iii) Preservation of green areas and incentives for green wedges will enhance ecosystem quality and biodiversity within the municipality		(iii) mapping to control green areas
		(iv) The increase of green infrastructure and green recreational areas can provide ecosystem services and function as nature-based solutions which can improve climate change adaptation		

CIVIL SOCIETY AND A VIBRANT CITY

SO VIII: Transport	(i) The increase of public transport and bike lanes, as well as the maintenance of it, can risk disturbing ecosystems, create noise pollution, and interfere with other land-use policies (demands land)	(i) An increase and maintenance of sustainable public transport options will increase ridership and decrease GHG emission	(i) Swedish Transport Administration (Trafikverket). Environments- and Spatial planning authority.	(i) Measure carbon dioxide emissions. (ii) Measure traffic noise levels, and number of screens.
	(ii) Increase of traffic risks between bike and car/bus traffic if not planned correctly	(ii) With increased mobility and transport services the efficiency and accessibility within the city will increase	(ii) Sweden's Climate Act and Climate Policy Framework. Naturvårdsverket.	(iii) Measure length of transportation infrastructure contra ecological corridors and green wedges.
	(iii) Soil consumption	(iii) With an increase of the transportation network and its reach, the improved accessibility of the city will provide more equitable mobility and accessibility conditions for all citizens		(iv) GIS-mapping.
		(iv) Increase bike lanes (and the needed bike infrastructure) will increase bike ridership → less emissions		
SO IX: Housing	(i) Repurposing and investment in renovating and reusing old buildings for housing can lead to higher accommodation prices within these areas, and in some cases fuel gentrification and segregation trends	(i) Mix-use development in line with transit corridors can increase public transport ridership and decrease car dependency, and associated emissions.	(i) Samhällsbyggnadsnämnden (Spatial Planning Authority)	(i) Average housing price, income level, and rental to ownership by neighborhood.
	(ii) The repurposing and mix-use development within the city can lead to significant risks for the cultural heritage of the city	(ii) The incorporation of mix-use development/repurposing can further spur local activity and safety within the communities	(ii) Planning and building act	(ii) Number of annual mixed used conversions.
	(iii) With mix-used planning objectives risks increasing the demand for land and industry spaces which can fuel further soil consumption and create conflicts between housing development, green parks and recreational areas objectives	(iii) Interlinked to Strategic Option 4, the facilitation of mix-use development can increase the collaboration between communities, local businesses, universities and industries.		(iii) Mapping to follow housing density. Socio economic index to follow segregation trends.
		(iv) With the decrease of agricultural soil consumption, due to decreased housing developments, the ecosystems outside of the city center will be better preserved		(iv) For soil, identify contaminated land, among industrial land use intensity.

Table 6: Monitoring actions

4. Discussion and conclusion

4.1 Discussion recommendations

Linköpings Municipality sets clear ambitions to move towards a sustainable direction. The municipality focuses on concentration and development. A more united urban form, a circle rather than a banana-shaped city, aligns well with presented sustainability goals, especially the ones related to reducing transport demand. Furthermore, the plan to reshape the municipality aligns with efficient land use, and climate mitigation. Although, the current comprehensive plan dates back to 2010, the ongoing development of a new digital comprehensive plan shows institutional awareness. This awareness indicates the need for an updated and integrated plan. However, the effectiveness of this transition relies on well-defined sustainability goals and its objectives, and how these objectives are put into action. Another concern is how to implement these sustainable questions into regulatory frameworks. These defined sustainability questions should be systematically monitored over time. A noticeable challenge in strategic planning isn't formulating sustainability ambitions and plans, since the municipality has these well thought through. Rather, the challenge lies in translating them into legally detailed plans and procedures for long-term investment priorities. It was noticeable that Linköping suffers from a lack of recreational areas that can be reached without cars, housing affordability issues and public transportation outside the city center. The formulation of the municipal sustainable plan as a whole is not the problem, but translating these plans and integrating them with how people live today in linköping could be difficult. According to the final matrix and its legends table, affordable on campus student housing had the highest score, while sustainable industry had the lowest score. Some recommendations to achieve sustainable housing is to implement energy efficient buildings, green roofs and implement renewable energy. Furthermore, to reach sustainable green industry, outside of what lnköping already does, a recommendation could be to implement green areas within the industry, promoting ecological networking and turning grey areas into green.

A topic raised in the interviews was the development of Ostlänken. The project represents infrastructural investment aimed to support regional connectivity. While such infrastructure can contribute to shifting the climate goals from roads to rail, its spatial demand requires critical examination. Ostlänken lifts concerns about land consumption and fragmentation. To cooperate with existing infrastructure, such as existing roads, would be a recommendation for a sustainable land-use approach. This would likely reduce environmental impact and minimize changing of underdeveloped land. Furthermore, another recommendation would be to implement green corridors in the ecosystem to reduce fragmentation. Without such integrations, the project could undermine sustainability and the sustainable gains achieved through compact urban planning.

Another objective is the protection of agricultural land. However, the municipality argues that the usage of agricultural land is sometimes necessary to facilitate urban growth. Furthermore, another argument for using urban land is the need to improve connectivity within the municipality. This position creates a conflict of development and preservation. Agricultural land can function as an economic resource, strengthen biodiversity, and cultural values. A

recommendation is to address the increasing uncertainty regarding agricultural land, the long term societal value of farmland may outweigh short term urban expansion. It is therefore recommended to plan strategies that apply with a stronger justification criteria regarding land use solutions before approving development on agricultural land.

At the detailed planning level, green infrastructure, ecosystem services, and accessible recreational environments should be emphasized. The limitation to sanitary water bodies and beaches without a car suggest spatial imbalance. Spatial planning should not only benefit land-use, but also address socio-spatial disagreements within the land-use sector. By ensuring mobility across demographic groups and recreational areas would enhance social well being. Furthermore, addressing housing affordability regularly, demographic trends, and market dynamics could prevent socioeconomic segregation. Without continuous evaluation, there is a risk that well designed strategies risk losing effectiveness over time. Additionally, to address the social segregation in the land use sector of linköping, social sustainability requires close attention. The perceived separation between student and “ordinary” city life suggests social fragmentation. Better integration of student housing is therefore a recommendation.

In conclusion, Linköping provides and demonstrates a strong sustainable and strategic planning. However, several gaps between growth ambitions, infrastructure, and social unity are remaining. We believe that Linköping would benefit by implementing regulatory controls over spatial societal fragmentations, create stronger planning that justifies the usage of agricultural land, and plan for recreational accessibility without the usage of cars. Furthermore, industries could benefit the environment, instead of just acting as a source of pollution, by implementing buffer zones and creating green areas within industry land. Affordable student housing demands would create the opportunity to implement green roofs, energy efficient buildings and place housing closer to recreational areas and cities to remove isolation between students and "normal" city life.

4.2 Uncertainties and gaps

Despite the structured approach of the SEA, several uncertainties remain that should be addressed in future research to strengthen the robustness of the assessment.

One key uncertainty concerns the spatial implications of compact development and agricultural land protection. While the preferred strategy emphasises inward growth and reduced soil consumption, no quantitative spatial modelling was undertaken to calculate projected land-take under different scenarios or to assess impacts on high-quality agricultural soils and ecological connectivity. Future GIS-based scenario analysis would strengthen confidence that the selected strategy minimises irreversible land transformation.

For SO VI (affordable student housing) and SO IX (housing), both of which rely on compact development principles, future spatial capacity analyses could provide greater clarity on how projected housing demand aligns with available land within campus areas and transit corridors. More detailed modelling of infrastructure capacity and land-use efficiency would help optimize density levels while ensuring environmental performance and agricultural land protection goals are maintained.

For SO III, the integration of Industry functions within mixed-use environments presents strong innovation potential; however, further assessment of regulatory adaptability, market demand, and infrastructure compatibility would refine the operational framework of this strategy. Such analysis would support effective zoning adjustments and ensure long-term economic viability.

Regarding SOII, the activation of cultural tourism through interactive heritage pathways offers significant socio-economic benefits. Future studies focusing on visitor management, maintenance planning, and heritage conservation thresholds would help balance economic stimulation with long-term preservation objectives.

While the recommended strategic options remain robust within the Strategic Thinking SEA framework, their implementation would benefit from complementary technical assessments tailored to each pathway. Such additional technical evaluations would not alter the strategic direction but would strengthen its effectiveness, scalability, and long-term sustainability.

4.3 Conclusion

This Strategic Environmental Assessment (SEA) evaluated the land-use sector of Linköping Municipality's comprehensive plan. The primary aim was to guide the city's long-term development toward becoming a "vibrant and attractive city, while respecting nature and the historical heritage of the town". As one of Sweden's fastest-growing cities, Linköping faces the complex challenge of accommodating a projected population of up to 250,000 residents while simultaneously preserving valuable agricultural land, managing the spatial demands of the Ostlänken railway, and mitigating socio-spatial segregation.

Through a Strategic Thinking SEA framework, the assessment identified Critical Decision Factors (CDFs) and evaluated nine distinct strategic options. Based on the risk-opportunity matrix, the following strategic pathways are highly recommended to optimize sustainable growth:

- **Affordable on-campus student housing (SO VI):** This option scored the highest in the assessment and directly addresses the urgent need for affordable accommodations, which will help bridge the social isolation between university students and the broader city population.
- **Cultural tourism (SO II):** Connecting historical sites through interactive pathways will enhance the city's cultural identity and stimulate the local economy.
- **Innovative industry (SO III):** Integrating Industry 5.0 productive uses into the ground floors of mixed-use buildings will foster innovation and reduce the need for external industrial sprawl.
- **Mixed-use housing development (SO IX):** Concentrating housing expansion along transit corridors will minimize the irreversible consumption of agricultural land.

While Linköping demonstrates strong environmental ambitions, the ultimate challenge lies in translating these high-level goals into legally detailed implementation plans. Moving forward, the municipality must rely on continuous spatial monitoring via Geographic Information Systems (GIS) to evaluate environmental impacts over time. By integrating stricter criteria for agricultural land use, expanding green corridors, and improving public

transit connectivity outside the city center, Linköping can successfully balance its rapid urbanization with long-term ecological and social resilience.

5. References

- Adamiak, C. & Marjavaara, R. (2024). Airbnb and urban population change: an empirical analysis of the case of Stockholm, Sweden. *Urban Research and Practice* 17(5). Routledge: Taylor & Francis Group.
- Azcárate, J. (2026, January 22). Strategic thinking for sustainability (ST4S) (Lecture 2). KTH Royal Institute of Technology
- Balfors, B. (2026). Workshop 5: Opportunities/Risk assessment, monitoring and follow-up. *KTH Royal Institute of Technology*. Slides: 7-8.
- Bellander, G. (n.d.). Blandstaden ett planeringskoncept för en hållbar bebyggelseutveckling?. *Boverket*. (Online). Available at: <https://www.boverket.se/globalassets/publikationer/dokument/2005/blandstaden.pdf> (Accessed: 18 Feb 2026)
- Birch, K. (2016). Materiality and sustainability transitions: integrating climate change in transport infrastructure in Ontario, Canada. *Prometheus* Vol.34. Routledge: Taylor & Francis Group.
- Boverket. (2015). Industri- och annat verksamhetsbullen vid planläggning och bygglovsprövning av bostäder. *Boverket*. (Online). Available at: <https://www.boverket.se/globalassets/publikationer/dokument/2015/industri-och-annat-verksamhetsbullen-vid-planlaggning-och-bygglovsprovning-av-bostader2.pdf> (Accessed: 18 Feb 2026)
- Boverket (2023). Energy performance certificate. (Online). Available at: <https://www.boverket.se/en/start/laws-and-regulations/national-regulations/energy-performance-certificate/> Fetched 2026-02-21 (Accessed: 17 Feb 2026)
- Boverket. (2024). Funktionsblandning på rätt sätt. *Boverket*. (Online). Available at: https://www.boverket.se/sv/PBL-kunskapsbanken/oversiktsplan/allmanna-intressen/hansyn/miljo_klimat/klimatpaverkan/klimatsmarta-strukturer/funktionsbland/ (Accessed: 19 Feb 2026)
- Campo-Ruiz, I. (2023). Economic powers encompass the largest cultural buildings: market, culture and equality in Stockholm Sweden (1918-2023). *International Journal of Architectural Research* 19(1). Emerald Publishing Limited.
- Caneloro, G. & Tartari, M. (2025). Heritage-led sustainable development in rural areas: the case of Vivi Calascio community based cooperative. *Cities* Vol.161. Elsevier.
- County Administrative board of Östergötland. (n.d) Länsstyrelsen.(Online)). Available at: (n.d)<https://www.lansstyrelsen.se/english/about-us/the-county-administrative-board.html> (Accessed: 18 Feb 2026)
- Council of Europe Portal. (2026). Cultural Routes of the Council of Europe Programme. *Council of Europe Portal*. (Online). Available at: <https://www.coe.int/en/web/cultural-routes/about>(Accessed: 18 Feb 2026)
- Cyprianska, M. & Nezlek, J. (2018). Everyone can be a winner: the benefits of competing in organized races for recreational runners. *The Journal of Positive Psychology*. Taylor & Francis.
- Di Figlia, L. (2016). Turnaround: abandoned villages, from discharged elements of modern Italian society to possible resources. *International Planning Studies* 21(3). Routledge: Taylor & Francis Group

EEA. (2024). How does environmental noise pollution impact my health?. European Environmental Agency. (Online). Available at: <https://www.eea.europa.eu/en/about/contact-us/faqs/how-does-environmental-noise-pollution-impact-my-health> (Accessed: 5 Oct 2025)

Esmail, B., Cortinovis, C., Suleiman, L., Albert, C., Geneletti, D. & Mörtberg, U. (2022). Greening cities through urban planning: A literature review on the uptake of concepts and methods in Stockholm. *Urban Forestry & Urban Greening* Vol.72. Elsevier

European Commission. (2023). *Strategic Environmental Assessment*. Capacity4dev. (Online). Available at: https://capacity4dev.europa.eu/groups/public-environment-climate/info/strategic-environmental-assessment_en (Accessed: 18 Feb 2026)

European Commission. (n.d.). SE_SEA_summary_report.pdf (Version 1.0) (PDF). (Online). Available at: <https://circabc.europa.eu/ui/group/3b48eff1-b955-423f-9086-0d85ad1c5879/library/879f1afe-9261-4353-afb3-7c97319db6cc/details?download=true> (Accessed: 18 Feb 2026)

European Commission. (2000). Directive 2000/60/EC Establishing framework for community action in the field of water policy.(Online). Available at: <https://eur-lex.europa.eu/legal-content/EN/ALL/?uri=celex:32000L0060> (Accessed: 12 Feb 2026)

European Commission. (2001). *Directive 2001/42/EC on the assessment of the effects of certain plans and programmes on the environment (SEA Directive)*. (Online). Available at: <https://eur-lex.europa.eu/eli/dir/2001/42/oj/eng> (Accessed: 12 Feb 2026)

European Parliament & Council of the European Union. (2018, May 30). Regulation (EU) 2018/841 of the European Parliament and of the Council of 30 May 2018 on the inclusion of greenhouse gas emissions and removals from land use, land-use change and forestry in the 2030 climate and energy framework, and amending Regulation (EU) No 525/2013 and Decision No 529/2013/EU (consolidated version as of 11 May 2023). EUR-Lex. (Online). Available at: <https://eur-lex.europa.eu/legal-content/EN/TXT/?uri=CELEX:02018R0841-20230511> (Accessed: 18 Feb 2026)

Eurocities. (2022, December 1). *Linköping - Eurocities. Eurocities - Home*. (Online). Available at: <https://eurocities.eu/cities/linkoping/#:~:text=Innovation%20and%20sustainability%20reflects%20the,in%20several%20high%20technology%20areas>. (Accessed: 18 Feb 2026)

a.European Union. (n.d.). Industry 5.0. *European Union*. (Online). Available at: https://research-and-innovation.ec.europa.eu/research-area/industrial-research-and-innovation/industry-50_en (Accessed: 18 Feb 2026)

b.European Union. (n.d.). Industrin och den gröna given. *European Union*. (Online). Available at: https://commission.europa.eu/topics/business-and-industry/industry-and-green-deal_sv (Accessed: 18 Feb 2026)

European Union. (2023). *Legal context for Strategic Environmental Assessment in Sweden*. (PDF). CIRCABC. (Online). Available at: <https://circabc.europa.eu/ui/group/3b48eff1-b955-423f-9086-0d85ad1c5879/library/879f1afe-9261-4353-afb3-7c97319db6cc/details?download=true> (Accessed: 18 Feb 2026)

Femenias, P., Peter, S. & Legner, M. (2023). Modern heritage and housing renovation: Policy development and practical experiences from Gothenburg, Sweden. *IN SITU-REVUE DE PATRIMOINES* Vol.49.

Glaeser, E.L. & Kahn, M.E. (2004). Chapter 56: Sprawl and Urban Growth. *Handbook of Regional and Urban Economics* Vol.4. Elsevier.

Greengo Energy. (n.d.). Bjursby solcellspark. *Greengo Energy*. (Online). Available at: <https://www.greengoenergy.se/bjursby-solcellspark> (Accessed: 18 Feb 2026)

Hassan, Q., Algburi, S., Sameen, A.Z., Salman, H.M. & Jaszczur, M. (2023). A review of hybrid renewable energy systems: solar and wind-powered solutions: challenges, opportunities, and policy implications. *Results in Engineering* Vol. 20. Elsevier.

Heimstaden. (n.d.). Lediga bostäder- Linköping. *Heimstaden*. (Online). Available at: https://heimstaden.com/se/sok-lagenhet/?text=Link%C3%B6ping&display_mode=picture&search_version=1.5&order=&tracking_object_type=Apartment&google_place_id=&google_place_type=&google_place_name=&number_of_rooms_min=1&number_of_rooms_max=10&rent_min=0&rent_max=26000&size_min=0&size_max=300&properties_true_false%5B%5D=student&offset=15&map_center=58.408492972175665,15.56544515212801&map_zoom=15.711223602294922 (Accessed: 18 Feb 2026)

Hupkes, J. (2024). A changing utopia: a critical analysis of a heritage of change in a 20th-century single-family housing area of cultural significance. *Bebyggelsehistorisk Tidskrift* Vol.86. Art & Architecture Complete.

IBEF. (2026). Landsbygdsturism i Indien: Stärka byar och återuppliva lokala ekonomier. *IBEF*. (Online). Available at: <https://www-ibef-org.translate.goog/blogs/rural-tourism-in-india-empowering-villages-and-reviving-local-economies? x tr sl=en& x tr tl=sv& x tr hl=sv& x tr pto=sge> (Accessed: 18 Feb 2026)

Ingvardsson, H.B. & Nielsen, O.A. (2018). How urban density, network typology, and socio-economy influence public transport ridership: Empirical evidence from 48 European metropolitan areas. *Journal of Transport Geography* Vol.72. Elsevier.

Jones, A. (2025). Reclaiming urban spaces: how cities are welcoming nature back. *European Union*. (Online). Available at: <https://projects.research-and-innovation.ec.europa.eu/en/horizon-magazine/reclaiming-urban-spaces-how-cities-are-welcoming-nature-back> (Accessed: 19 Feb 2026)

Krisinformation. (2023). Sweden's climate goals. *Krisinformation*. (Online). Available at: <https://www.krisinformation.se/en/hazards-and-risks/climate-change/swedens-climate-goals/> (Accessed: 17 Feb 2026)

Land use sector. (n.d.). Climate Action. (Online). Available at: https://climate.ec.europa.eu/eu-action/land-use-sector_en (Accessed: 17 Feb 2026)

a.LiU. (n.d.). Så gör LiU nytta i samhället. *Linköpings Universitet*. (Online). Available at: <https://liu.se/samverkan/sa-gor-liu-nytta-i-samhallet> (Accessed: 17 Feb 2026)

b.LiU. (n.d.). Collaboration at Linköping University. *Linköping University*. (Online). Available at: <https://liu.se/en/collaboration> (Accessed: 18 Feb 2026)

c.LiU. (n.d.). Walk the Talk- Energismarta och hållbara fastigheter. *Linköpings Universitet*. (Online). Available at: <https://liu.se/artikel/walkthetalk-energismarta-fastigheter> (Accessed: 18 Feb 2026)

Linköping kommun. (2010). *Översiktsplan för staden Linköping: Linköpings översiktsplan del 2*. Larsson Offsettryck AB.

Linköpings Kommun. (2025). Investeringsguide till Linköpings företagsområden. *Linköpings Kommun*. (Online). Available at: <https://www.linkoping.se/framtidsstaden/etablera-och-investera/investeringsguide-till-linkopingsforetagsomraden> (Accessed: 18 Feb 2026)

Linköping University. (2020). *Magnus Klofsten – professor of bright ideas*. (Online). Available at: <https://liu.se/en/news-item/magnus-klofsten---professor-i-lysande-ideer> (Accessed: 18 Feb 2026)

Länsstyrelsen Östergötland. (n.d.). Byggnadsminnen. *Länsstyrelsen Östergötland*. (Online). Available at: <https://www.lansstyrelsen.se/ostergotland/samhalle/kulturmiljo/byggnadsminnen.html> (Accessed: 17 Feb 2026)

Länsstyrelsen Östergötland. (n.d.). Tinnerö eklandskap. (Online). Available at: <https://www.lansstyrelsen.se/ostergotland/besoksmal/naturreservat/tinnero-eklandskap-naturreservat.html> (Accessed: 18 Feb 2026)

Maguire, D. J., Batty, M., & Goodchild, M. F. (2005). *GIS, spatial analysis, and modeling*. In D. J. Maguire, M. Batty, & M. F. Goodchild (Eds.), *GIS, spatial analysis, and modeling* (pp. 1–12). ESRI Press. (Online). Available at: https://www.geos.ed.ac.uk/~gisteac/gis_book_abridged/files/ch68.pdf (Accessed: 18 Feb 2026)

Monaco, S., Freire Varela, A., Prospero, C. & Corbisiero, F. (2026). Social entrepreneurship as a driver for social inclusion and sustainable tourism: the case of Herculaneum, Italy. *Cities* Vol.171. Elsevier.

Naturvårdsverket. (2023). Sweden's climate act and climate policy framework. *Naturvårdsverket*. (Online). Available at: <https://www.naturvardsverket.se/en/international/swedish-environmental-work/swedens-climate-act-and-climate-policy-framework/> (Accessed: 19 Feb 2026)

Naturvårdsverket. (2025). Industrin, utsläpp av växthusgaser. *Naturvårdsverket*. (Online). Available at: <https://www.naturvardsverket.se/data-och-statistik/klimat/vaxthusgaser-utslapp-fran-industrin/> (Accessed: 18 Feb 2026)

OECD. (n.d.). Knowledge exchange and co-creation in European Union. *OECD*. (Online). Available at: <https://stip.oecd.org/stip/interactive-dashboards/countries/EuropeanUnion/themes/TH5> (Accessed: 17 Feb 2026)

Ortega, J.A., Asfour, O.S. & Wuni, I.Y. (2026). A systematic review of renewable energy integration in social housing and its impacts on affordability and community sustainability. *Discover Sustainability* 7(108). Creative Commons Attribution.

Partidário, M.R. (2012). *Strategic Environmental Assessment Better Practice Guide: Methodological Thinking for Strategic Thinking in SEA*. Portuguese Environment Agency & Redes Energéticas Nacionais (REN).

Peiser, R. (2001). Decomposing urban sprawl. *Town Planning Review* 72(3). Liverpool University Press.

Regeringskansliet. (n.d.). Acceleration för industrins omställning och konkurrenskraft. *Regeringskansliet*. (Online). Available at: <https://www.regeringen.se/regeringens-politik/acceleration-for-industrins-grona-omstallning-och-konkurrenskraft/> (Accessed: 18 Feb 2026)

Region Stockholm. (2025). Grönstruktur. *Region Stockholm*. (Online). Available at: <https://www.regionstockholm.se/regional-utveckling/regional-utvecklingsplanering/fysisk-regionplanering/gron-och-h-blastruktur/> (Accessed: 19 Feb 2026)

Riksdag (2010). Plan - och bygglag (2010:900). (Online). Available at: https://www.riksdagen.se/sv/dokument-och-lagar/dokument/svensk-forfattningssamling/plan-och-bygglag-2010900_sfs-2010-900/ (Accessed: 18 Feb 2026)

Sadler, B., Aschemann, R., Dusik, J., Fischer, T., Partidário, M. R., & Verheem, R. (2011). *Handbook of Strategic Environmental Assessment*. Earthscan.

Samant, S.R. (2026). Regenerating industrial neighbourhoods with systems thinking and circularity: Sungei Kadut, Singapore. *Journal of Regeneration and Renewal* 19(2). Henry Stewart Publications LLP.

Slocum, L.A., Greene, C., Huebner, B.M., Udani, A. & Spanoudakis, K. (2024). Centering community in conceptualization and measurement of safety: a mixed method approach. *Justice Quarterly* 41(7). Routledge: Taylor & Francis Group.

SPAB. (n.d.). Reusing old buildings. *SPAB*. (Online). Available at: <https://www.spab.org.uk/advice/reusing-old-buildings> (Accessed: 18 Feb 2026)

Stockholm University. (n.d.). Student Unions. *Stockholm University*. (Online). Available at: <https://www.su.se/english/education/student-life/student-unions> (Accessed: 18 Feb 2026)

Studentbostäder. (n.d.). Lediga bostäder just nu. *Studentbostäder Linköping*. (Online). Available at: <https://www.studentbostader.se/> (Accessed: 18 Feb 2026)

a.Sveriges Riksdag. (1998). *Miljöbalk. (1998:808)* Sveriges Riksdag. (Online). Available at: https://www.riksdagen.se/sv/dokument-och-lagar/dokument/svensk-forfattningssamling/miljobalk-1998808_sfs-1998-808/ (Accessed: 18 Feb 2026)

b.Sveriges Riksdag. (1998). *Kulturmiljöbalk. (1998:950)* Sveriges Riksdag. (Online). Available at: https://www.riksdagen.se/sv/dokument-och-lagar/dokument/svensk-forfattningssamling/kulturmiljolag-1988950_sfs-1988-950/ (Accessed: 18 Feb 2026)

Sveriges Riksdag (2010). Plan - och bygglag (2010:900). Sveriges Riksdag (Online). Available at: https://www.riksdagen.se/sv/dokument-och-lagar/dokument/svensk-forfattningssamling/plan-och-bygglag-2010900_sfs-2010-900/ (Accessed: 18 Feb 2026)

Sweden Green Building Council. (n.d.). Miljöbyggnad- Sveriges ledande certifiering för hållbar ny- och ombyggnation. *Sweden Green Building Council*. (Online). Available at: <https://www.sgbc.se/certifiering/miljobyggnad/> (Accessed: 18 Feb 2026)

Swedish Environmental Protection Agency, 2023. *Miljöbedömning av planer och program*. (Online). Available at: <https://www.naturvardsverket.se/vagledning-och-stod/miljobalken/miljobedomningar/strategisk-miljobedomning/> (Accessed: 18 Feb 2026)

Szulcewska, B., Giedych, R., Borowski, J., Kuchcik, M., Sikorski, P., Mazurkiewicz, A. & Stanczyk, T. (2014). How much green is needed for a vital neighbourhood? In search for empirical evidence. *Land Use Policy* Vol.38. Elsevier.

Tekniska Verken. (n.d.). Solcellsparken i Linköping- en av de största i Sverige. *Tekniska Verken*. (Online). Available at: <https://www.tekniskaverken.se/om-oss/vad-vi-gor/solkraft/solcellsparken-linkoping/> (Accessed: 18 Feb 2026)

Therivel, R. (2010). *Strategic Environmental Assessment in Action* 2nd Edition. Earthscan by Routledge.

Thörn, H., Sernhede, O., & Lindgren, H. (2023). Ansvarets politik: Kommunledd gentrifiering och hyrestgästorganisering. *Sociologisk Forskning*, 60(1)

Tillväxtverket. (2014). Fakta om svensk turism: turismens effekter på ekonomi, export och sysselsättning samt volymer, beteenden, utbud och efterfrågan. *Tillväxtverket*. (Online). Available at: https://tillvaxtverket.se/download/18.6855bfcf184896002ff894/1668765480052/info_0613_web.pdf (Accessed: 18 Feb 2026)

Tillväxtverket. (2026). Bidra till en grön omställning av näringslivet. *Tillväxtverket*. (Online). Available at: <https://tillvaxtverket.se/tillvaxtverket/sokfinansiering/utlysningar/utlysningar/bidratillengronomstallningavnaringslivet.10827.html> (Accessed: 18 Feb 2026)

Trafikverket. (2025). *Ostlänken – en ny dubbelspårig järnväg*. (Online). Available at: <https://www.trafikverket.se/vara-projekt/projekt-som-stracker-sig-over-flera-lan/ostlanken-en-ny-dubbelsparig-jarnvag/> (Accessed: 15 Feb 2026)

Tunström, M. and Wang, S. (2019) The segregated city. *Nordic Council of Ministers*. (Online). Available at: <https://doi.org/10.6027/Nord2019-007> (Accessed: 19 Feb 2026)

Turbay, A., Pereira, R., & Firmino, R. (2024). The equity implications of TOD in Curitiba. *Case Studies on Transport Policy* Vol. 16. Elsevier Ltd.

United Nations (n.d.). The 17 Goals. *United Nations: Department of economic and social affairs*. (Online). Available at: <https://sdgs.un.org/goals> (Accessed: 17 Feb 2026)

UNECE. 2023. *Strategic Environmental Assessment at the Policy Level: Recent Progress, Current Status and Future Prospects*. (Online). Available at: https://unece.org/fileadmin/DAM/env/eia/documents/PolicySEA/SEA_of_Policies_volume.pdf (Accessed: 18 Feb 2026)

UNESCO. (2024). Student unions: ensuring the right to quality and accessible education for all. *UNESCO*. (Online). Available at: <https://www.unesco.org/sdg4education2030/en/articles/student-unions-ensuring-right-quality-and-accessible-education-all> (Accessed: 18 Feb 2026)

United Nation Development Program, u.d. *The SDGs in Action*. (Online). Available at: <https://www.undp.org/sustainable-development-goals> (Accessed: 18 Feb 2026)

Vanhöfen, N., Härtel, T., Reichert, G. and Radler, C. (2025) ‘The relationship between perception and landscape characteristics of recreational places with human mental well-being’, *Scientific Reports*, 15.

Visit Linköping. (n.d.). Upplev kulturen i Linköping. *Visit Linköping*. (Online). Available at: <https://visitlinkoping.se/se-och-gora/kultur-i-linkoping/> (Accessed: 18 Feb 2026)

Visit Sweden. (n.d.). Kulturturism. *Visit Sweden*. (Online). Available at: <https://corporate.visitsweden.com/kunskap/kulturturism/> (Accessed: 18 Feb 2026)

Vulpiani, M., Rossetti, C., Caiffa, M. & Petrongolo, S.C.L. (2026). The importance of assessing the economic and social value of cultural heritage: the colosseum. *Journal of Cultural Heritage Management and Sustainable Development* 16(7). Emerald Publishing Limited.

Wimark, T., Andersson, E.K. & Malmberg, B. (2021). Bostadssegregation i Södertörn 1990-2012: Förtätning och ombildning med olika utfall. In: Att göra stad i Stockholms urbana periferi. Gawell, M. & Apostolis, T. Stockholm: Stockholmia förlag.

Xu, X. and Lan, Y., 2016. A comparative study on carbon footprints between plant-and animal-based foods in China. *Journal of Cleaner Production*, 112, pp.2581-2592.

Zhang, A., Wharton, C., Cloonan, S., Bronowsky, R., Magesh, V., Braverman, I., Marquez, A., Leidy, H., Wang, M.C., Cleveland, D.A., Jay, J. & Stecher, C. (2024). Changing the default meal option at university events to reduce harmful environmental impacts: six randomized controlled trials. *Appetite* Vol.200. Elsevier.

6. List of Figures

Figure 1: Insertion of the city of Linköping within the general framework of Östergötland County, which in turn is situated within the mosaic of Swedish counties. Source: authors elaboration through Adobe photoshop. Google satellite base.	5
Figure 2: Linköping Resecentrum (Travel Center)	6
Figure 3: Planned Ostlänken (The East Link) from Järna to Linköping -2035 (Trafikverket, n.d.)	6
Figure 4: EIA Based SEA process Source: (Azcárate, 2026)	10
Figure 5: Flow diagram showing Strategic thinking SEA process in planning	11
Figure 6: Flow diagram showing Intended analytical framework guiding the SEA of the Comprehensive Plan of Linköping. Source: Authors, adapted from (Azcárate, 2026)	24
Figure 7: Scope of master plan Source: Linköping kommun. (2010)	25
Figure 8: Stakeholders and main actors	27
Figure 9: Strategic options	40

7. List of Tables

Table 1: SWOT Analysis	19
Table 2: Cross-SWOT Analysis	21
Table 3: Ranking rubrics	27
Table 4: CDFs and assessment criteria	33
Table 5: Final matrix and its legends	51
Table 6: Monitoring actions	55

Appendices

A - Inhabitants interview (Anna Brobeck & Linnea Ingren)

For the purpose of transparency, the sentences are taken from what was being said and not rewritten. However, for a better understanding, grammar, spelling, and punctuation have been corrected.

Interviewer: Can you introduce yourself in a couple of sentences?

Anna: Ok, so my name is Anna. I'm 23 years old. I studied urban planning at Linköping University, and I graduated from my bachelor's there last year.

Linnea: I'm Linnea. I'm 24 years old. I studied in the same classes as Anna, so I also studied urban planning at Linköping University. I've lived in Linköping for three years, and I'm still a student.

Interviewer: Why did you decide to move and live in Linköping?

Anna: I think I chose Linköping because it has a very good student life and a big university with many good programs to study. Also, Linköping is one of the few universities in Sweden that has the program we studied. I grew up in Norrköping, which is close to Linköping, so Linköping has always been close, and that was a natural step for me to move there.

Linnea: Yeah, and I've heard a lot of good things about Linköping as a university city from people I know. Also, it's pretty close to Stockholm, where I grew up, so it felt pretty natural because I wanted to study in a big university city. It felt more natural than going to Lund or Umeå, for example. I also heard a lot of good things about the program I was studying.

Interviewer: You studied urban planning and design?

Linnea and Anna: No, it was urban planning.

Anna: More like urban and regional planning.

Linnea: Yes, maybe it's called urban and regional planning.

Interviewer: Would you recommend living here?

Anna: Yes, I would. It's a very nice city that has a lot of services. You have rich nature, and it's easy to get around the whole city by bike. As a student, living there was the perfect city, I feel like.

Linnea: Yeah, I wouldn't change that, so I would really recommend studying there. But I don't know if I would want to live there for a long time. Maybe for a shorter time. Compared to other cities, I would prefer to live in a bigger city.

Interviewer: Thanks for your comments on Linköping. Do you think housing, especially for students and adults, is affordable for people working there?

Anna: I think they have built a lot of new housing in the central parts of Linköping, but the problem today is that they are very expensive. All of the smaller apartments that students want to live in are hard to find and very expensive. That's the main problem. There is a lot of housing, but it's hard to find something affordable.

Linnea: Yeah, I agree with Anna. The problem isn't that there is too little housing. They build a lot, but as a student it's pretty expensive. I feel like I pay almost as much as I pay today in central Stockholm. In Linköping I lived in a smaller apartment, 20 m², and it took 40–45 minutes to walk to the city center. So it was pretty much outside the city, and still expensive. If you want your own apartment, it's expensive. Otherwise, you have corridor rooms.

Interviewer: Do students usually live in private apartments or university buildings?

Anna: Mostly private apartments.

Linnea: Yeah, but it's pretty common to share. In my experience in Linköping, people often share apartments.

Interviewer: But they belong to private owners?

Linnea: Yeah, exactly. It's communal.

Anna: They have Stångåstaden, which is a big company.

Linnea: Yeah, it's like a property company owned by the municipality.

Interviewer: Like SSSB?

Anna and Linnea: Yeah, exactly. Same concept.

Linnea: That's the most common.

Anna: The university doesn't have apartments, but the municipality does.

Interviewer: Are these apartments close to the university?

Anna and Linnea: Yeah.

Interviewer: But there are few?

Linnea: No, there are a lot, but they are pretty expensive, apart from corridor rooms. Corridor rooms are fine in price, I think.

Anna: If you look at Linköping on a map, the university is outside the city center. Almost all students live near the university. There's like a whole world of students. When we lived in Linköping, we both lived close to the university, so we weren't that much in the city.

Interviewer: Did you feel isolated?

Linnea: Yeah, actually. It felt like one world of students and one world of working people. I could go days or weeks without meeting anyone other than students. This is because students use only students' facilities, such as grocery shops, gyms...

Anna: People I know who didn't study at Linköping University but lived in Linköping didn't even recognize that there was a big university in the city.

Interviewer: What makes Linköping stand out compared to cities like Stockholm?

Linnea: The university is a big reason. It's a great university with a lot of good education.

Anna: There are also good opportunities for growth, with companies like SAAB and a strong technology sector.

Linnea: Yeah, research projects and new technology.

Anna: The city is very flat, with big farming fields around it.

Linnea: No hills at all.

Interviewer: Is it possible to enjoy the river?

Linnea: Yeah, but it's not very accessible from the university or city center.

Anna: There are walking paths, but the water isn't clean enough to swim in.

Linnea: If you want to swim, it's difficult.

Anna: You need a car or take the bus.

Interviewer: How do you spend your free time?

Linnea: We hang out with friends, have dinners, go partying, work out, and go for walks in the woods.

Anna: I like walking and running, especially in the Tinnerö Eklandskap nature reserve.

Interviewer: Would you consider Linköping a green city?

Anna: Yes, it's a green city with many parks and forests.

Linnea: Nature is easy to reach.

Interviewer: Do you see Linköping as a safe and vibrant city?

Anna: Yes, but areas like Skäggetorp and Berga feel less safe.

Linnea: You hear about shootings, but I haven't been there myself. Still, I wouldn't go there alone.

Anna: But I felt safe in the city. It is also very vibrant. There are people walking and going out every day, at every time. So, I always felt safe going home alone during the night.

Linnea: Me too.

Interviewer: Are there enough bars and clubs?

Linnea: There could be more. A lot is centered around the university.

Anna: There are many restaurants and bars, but less clubbing.

Interviewer: Do you think there are conflicts between those economic activities and residential areas? Are residents bothered by students' night life?

Anna: I think there is good communication between the owners and the ones living there. In some student areas where there's a club in the apartment building, I think that students are pretty much aware of the situation.

Linnea: But one thing I know is the area I lived in, called Vallastaden, close to the university. There are a lot of students who live there, but also some families, and I know that those families were quite annoyed by the students' noise at the time. Because sometimes there were parties in the middle of the week. So I feel like they, who live close to the university, have a more struggle with it than the people living in the city, hearing noise from the bars and so on.

Interviewer: How do you usually travel in Linköping?

Anna: Always by bike.

Linnea: Bike everywhere: Grocery shopping, meeting friends, school. After partying.

Anna: Even in winter. Sometimes we took the bus, but only with rain

Linnea: The bike lanes are very good. Bike paths are heated, so there's no ice or snow, so it's always easy to bike everywhere.

Anna: They are big and they're pretty much everywhere. But yeah, that's the easiest and fastest way to go around in the city. If you don't have the bike, you are quite locked in.

Linnea: Also because, as we said before, Linköping is very flat, so it's very easy to bike.

Interviewer: Even though you don't use public transportation, can you please give some comments on it? Like, is that accessible, affordable or efficient enough?

Anna: Yeah, I feel like it's very accessible and affordable as well. And they have a good thing where you get, like a discount on the ticket if you go by the transportation during peak time. So if you go in, like, in the middle of the day, there's a much better price. And you can get like 2 tickets for a ship or price if you're traveling with more than one person. So, yeah, I feel like it's very good and they're going on good times.

Linnea: I would say, I agree on the affordability part, but I, from where I live, a bit outside the city, I wouldn't actually say that it's so accessible. Like, in one direction it is very accessible, but I wasn't able to go by bus to school, and that's a 30 minute walk. And like, if I wanted to go to the shopping area, Toonville, where all the big stores were, there were like maybe one bus an hour for I need to like go to the city center and then change. So I feel they could have better bus options, actually. But it's very affordable.

Anna: They have good communication to other cities, and you go by the train called that is very fast. That's good, actually. And it's pretty affordable as well.

Interviewer: What makes Linköping sustainable?

Anna: New neighborhoods focus on resource efficiency and climate neutrality.

Linnea: I would say that transport is making the city sustainable. The bike culture is very strong. Of course, people have cars, but it's not so central.

Anna: They also focus on green energy and waste solutions.

Interviewer: What would you change?

Linnea: On the city level, I would improve the bus system. Earlier we talked about extending the bus network so that it doesn't only go to the city center and back. Of course, I understand that they can't extend lines if only ten people use them, but improving connectivity would be good. Apart from that, I think Linköping is a very good city to live in.

Anna: One noticeable thing is how the city is structured. The city center is one part, and then the family residential areas are more outside. The city has been described as growing in a "banana" shape. Now, there are plans to make it more "round" and better connected.

Linnea: Yes, right now people living at the ends of the "banana" are quite far apart. It would be better if the municipality grew more as a unit instead of spreading outward.

Interviewer: Thoughts on job opportunities?

Anna: Here are good opportunities, especially for engineers. Many companies want students to stay after graduation.

Linnea: There are many technology-oriented companies here, like SSAB. The city has ambitions to grow as a region focused on innovation and business. Compared to Stockholm, there are fewer job opportunities. That's common in Swedish cities outside Stockholm. Many people still move there for work.

Interviewer: During your studies at Linköping University, did you work on municipal planning projects?

Anna: Yes, we worked on projects related to new detailed development plans for future districts.

Linnea: For example, we studied Vallastaden, including its self-driving bus project. We had communication with the municipality and attended lectures about ongoing planning.

Interviewer: Are there any planning projects in Linköping that you find successful?

Linnea: Vallastaden is a good example.

Anna: Also the railway project Ostlänken. It aims to build faster regional trains. However, one issue is that the new station might be placed behind the current central station instead of directly in the center. That could reduce accessibility compared to Norrköping, where the railway will go directly into the central station.

Interviewer: What is the worst aspect of Linköping?

Linnea: It doesn't have a coastline. I miss being close to water and being able to swim easily in the summer. There is Roxen, which is very nice, but it's not very accessible. You often need a car, and public transport connections are limited. Making Roxen more accessible could be a good idea.

Interviewer: What about new developments?

Anna: The city is expanding with new neighborhoods. That's positive for development, but growth needs to stay connected.

Linnea: One challenge is the presence of military areas, such as Malmen. The military has veto power over municipal development, so some areas cannot be built on.

Interviewer: Is Linköping a historical city?

Anna: Not really. It feels more futuristic. The historical area is Gamla Linköping, with traditional small red wooden houses. There are also some historic buildings in the city center, like the cathedral.

Linnea: Some areas, like Skäggetorp, were built under the "Million Programme" model, where car traffic was kept

outside residential zones. These areas feel somewhat segregated. Other parts are industrial, including areas around the airport, which serves both military and civilian flights.

Interviewer: Is there a wealthy area in the city?

Anna: Not really within the city center. Wealthier residential areas are often low-density neighborhoods outside the core.

Linnea: Yeah, there isn't a clearly defined "rich district" like in larger cities.

Interviewer: Are there flooding risks?

Anna: Not significantly from the river. Flooding is more likely due to heavy rainfall and hard surfaces in the city.

Interviewer: Which sector should the city prioritize?

Linnea: Education and technology are already strong. Maybe transport and mobility could receive more focus.

Anna: I think the municipality is already doing a good job compared to others.

Interviewer: Is Linköping more a university city or an industrial city?

Linnea: More of a university and technology city, focused on innovation rather than traditional industry.

Interviewer: Any final thoughts?

Linnea: Linköping collaborates closely with Norrköping.

Anna: Together they want to become Sweden's fourth major region. There is a joint strategic plan guiding both municipalities. It's not legally binding, but it provides guidelines.

Interviewer: What about the relationship with Stockholm?

Linnea: Politically, they are separate regions. However, improved rail connections like Ostlänken will make commuting easier.

Anna: Many people already commute part-time, working in Stockholm but living in Linköping or Norrköping.

Interviewer: Thank you very much.

Anna and Linnea: Thank you so much for interviewing us. It was interesting!

B - Municipal comprehensive planner interview (Mikael Sonesson)

For the purpose of transparency, the sentences are directly pasted. However, for a better understanding, an English translation is added alongside in parentheses.

Question: Do you live in Linköping?

Mikael: Bor i Linköpings kommun, på landet 13 km från tätorten

(I live in Linköpings Municipality, in the countryside 13 km from the urban area)

Question: What is your role within the municipality?

Mikael: Översiktsplanerare

(Comprehensive planner)

Question: How long have you been working with Linköping municipality?

Mikael: 6 år

(6 years)

Question: What are the environmental protection plans

Mikael: Flera. Naturvårdsprogram, klimat- och energiprogram, klimatanpassning program, kemikalieprogram, åtgärdsprogram buller, åtgärdsprogram luftkvalitet, vattentjänst plan mm

(Several, Nature and conservation program, climate and energy program, climate adaptation program, chemical program, action plan for noise, action plan for air quality, water service plan and more)

Question: What makes Linköping stand out compared to different cities like Stockholm?

Mikael: Vet ej.

(I don't know)

Question: What does the land use sector look like right now in Linköping?

Mikael: Förstår inte frågan.

(I don't understand the question)

Question: What does Linköping hope to achieve with the land use sector?

Mikael: Få en rundare stad, möta behovet för 250 000 invånare på sikt.

(Create a more compact/round city and meet the needs of 250,00 inhabitants in the long term)

Question: In regard to Linköping's municipality's land-use sector, what challenges does the land use sector face?

Mikael: Hushållning med jordbruksmark, anpassning till ett förändrat klimat, Ostlänken, segregation mm.

(Responsible use of and agriculture, adaption to changing climate, the Ostlänken railway project, segregation and more)

Question: In regard to the overview plan for the city of Linköping 2010, have there occurred any obstacles?

Mikael: Ja, jag kan inte pekar på något särskilt. Det finns massa hinder hela tiden.

(Yes, I cannot point to anything specific. There are many obstacles all the time)

Question: Are you planning to make a new plan? Have the needs of the city changed in recent years?

Mikael: Vi håller på med en ny översiktsplan just nu. Nej.

(We are currently working on a new comprehensive plan. No)

Question: Have you included any SDGs, in that case which ones and why?

Mikael: Vi utgår från kommunens övergripande styrdokument. De globala hållbarhetsmålen ställer vi oss bakom i kommunens hållbarhetspolicy. Men de är svåra att göra operativa, de svenska miljömålen ligger mycket närmare vår verksamhet.

(We base our work on the municipality's overarching policy documents. We support the global sustainability goals in the municipality's sustainability policy. However, they are difficult to operationalize. The Swedish environmental objectives are much closer to our activities.)

Question: What are the municipality's objectives in relation to nature-based solutions and ecosystem protection?

Mikael: Vi har en riktlinjer för ekosystemtjänster och en för ekologisk kompensation. Vi eftersträvar öppna dagvattenlösningar som även ska ge ekosystemtjänster.

(We have guidelines for ecosystem services and one for ecological compensation. We aim for open stormwater solutions that also provide ecosystem services)

Question: What are your different land use categories?

Mikael: I den nya översiktsplanen är det funktionsblandad bebyggelse, tekniska anläggningar, verksamheter, grönområden och vattenområden.

(In the new comprehensive plan, the categories are mixed-use development, technical facilities, commercial industrial activities, green areas, and water areas.)

Question: What are your different land use plans and how do you apply them on different scales?

Mikael: Vi har just nu väldigt många fördjupningar och tematiska tillägg, de finns på kommunens hemsida. När vi antar den nya översiktsplanen kommer den att vara digital och ersätta samtliga befintliga delar av ÖP.

(We currently have many in-depth plans and thematic additions, they are available on the municipalities website. When we adopt the new comprehensive plan, it will be digital and replace all existing parts of the comprehensive plan)

Question: Is the conversion of agricultural land the most appropriate strategy for accommodating Linköping's urban expansion? What are the potential trade-offs in terms of food security, ecological impact, infrastructure costs, and long-term urban form? (refer to p 41 of the master plan)

Mikael: Vi vill undvika att ta jordbruksmark i anspråk, men för att tätorterna ska kunna växa är det ibland nödvändigt. Vi tar jordbruksmark i anspråk runt ett antal prioriterade orter och undviker en mer utspridd bebyggelse i landsbygdsområden, är strategin.

(We want to avoid using agricultural land, but for urban areas to grow it is sometimes necessary. Our strategy is to use agricultural land around a number of prioritized locations and avoid more scattered development in rural areas)

Question: How relevant do you think the 2010 master plan is in light of current conditions? What criteria should be prioritized based on current scenarios?

Mikael: Förstår inte frågan.

(I don't understand the question)

Question: Have the needs of the city changed in recent years? Is the municipality planning to amend the master plan, according to it?

Mikael: Mer fokus på klimatanpassning nu bl.a. Nytt läge för Ostlänken ger nya möjligheter och påverkar också, Ja ny ÖP är på gång.

(More focus on climate adaptation now, among other things. The new alignment of the Ostlänken railway creates new opportunities and also has an impact. Yes, a new comprehensive plan is underway)

Question: What makes Linköping a good example of sustainability?

Mikael: Vet inte

(I don't know)

Question: What should be changed here?

Mikael: Vet inte.

(I don't know)

Question: What are the major stakeholders that you have identified based on the land use sector from the master plan?

Mikael: Näringsliv, medborgare, civilsamhället, statliga myndigheter, kommunens egna organisationer.

(Business sector, citizens, civil society, government agencies, the municipality's own organizations)

Question: What are the conflicts that you have noticed while planning for the land use sector? One example might be- Is the conversion of agricultural land the most appropriate strategy for accommodating Linköping's urban expansion? What are the potential trade-offs in terms of food security, ecological impact, infrastructure costs, and long-term urban form? (refer to p 41 of the master plan)

Mikael: Jordbruksmark är en framkomlighet för olika trafikslag/ambitioner att få till en lägre andel bilresor en annan.

(Agricultural land is one. Accessibility for different modes of transport/ambitions to achieve a lower share of car travel is another)

Question: How are you balancing all the conflicting interests of the various stakeholders?

Mikael: Vi gör en professionell avvägning utifrån vår kompetens, den politiska viljan och det kommunala styrdokumentet.

(We make a professional assessment based on our expertise, the political will, and the municipal policy documents)

Question: Do you plan to include the locals in the planning process (and how)?

Mikael: Ja, längre fram. Via medborgardialoger.

(Yes, later on. Through citizen dialogues)

Question: Do you think there are other important points that were not touched upon during the interview?

Mikael: Nej

(No)